Carbon Price Transmission and Economic Protection: The Design of Border Carbon Adjustments

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Border carbon adjustments (BCAs) can be mass-based, charging all embodied emissions, or rate-based, charging emissions relative to a benchmark intensity. A rate-based BCA is the border counterpart of an intensity standard, which acts as an emissions tax plus an output subsidy: when applied to domestic production, the subsidy can limit leakage by keeping production at home; at the border, it instead supports foreign production. We develop a plant-level general equilibrium model of global steel and find that a rate-based BCA keeps most of the marginal price but removes the average charge. As a result, it does less for climate protection: a mass-based BCA would achieve the same global emissions at a third of the price in an EU setting and an eighth in a US setting. The average charge also determines who bears the economic burden: it raises fiscal revenue and protects domestic steelmakers, while its removal spares steel users. The rate-based design also shifts upstream production abroad, causing vertical leakage: EU ironmakers face carbon pricing while imported pig iron, a key intermediate input, is nearly uncharged. BCA design therefore fundamentally shapes climate effectiveness, protection for domestic producers, and the distribution of costs.

Border carbon adjustments (BCAs) are becoming real-world climate policy, and the European Union (EU) and the United States (US) are taking different routes. The EU's Carbon Border Adjustment Mechanism (CBAM) prices the embodied emissions of steel and other emissions-intensive imports at the EU carbon price, while US proposals would charge imports by how far their emissions intensity exceeds a US benchmark. The US route is not an outlier: where the domestic carbon price itself works through intensity benchmarks, as under free allocation in the EU until 2034, Canada's output-based pricing system, and China's national emissions trading

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system, a border measure that mirrors it also charges imports relative to a benchmark (ICAP, 2026). Behind the two policy routes lies a choice between charging for all emissions, as an emissions tax does, and charging only for emissions relative to a benchmark intensity, as an intensity standard does. We compare two designs: a mass-based BCA, which charges all embodied emissions, and a rate-based BCA, which charges emissions above a benchmark intensity and credits those below it. They differ in how they transmit a carbon price to foreign producers. A carbon price reaches a foreign producer in two ways: at the margin, as a reward for lower emissions intensity, and on average, as a charge on every tonne of embodied carbon that raises the price of its exports. A mass-based BCA transmits both; a rate-based BCA transmits the marginal price but removes the average charge: what exporters above the benchmark pay is credited to those below it. This paper asks how BCA design shapes climate effectiveness and economic incidence. How much global abatement does a rate-based BCA forgo when it retains strong incentives for cleaner production but removes the average charge on imports? How does this design choice affect fiscal revenue, protection for domestic producers, and costs for downstream users? We also examine whether it shifts upstream production abroad through trade in intermediates.

Within one economy, the choice between mass- and rate-based environmental policy instruments is well understood: an intensity standard acts as an emissions tax combined with an output subsidy, which makes it less efficient than a tax when all emitters are regulated (Helfand, 1991; Holland, Hughes and Knittel, 2009). The same subsidy makes output-based rebating, used in the free allocation of the EU Emissions Trading System (ETS), the most common remedy for leakage: it keeps regulated producers competitive against unregulated ones (Fowlie, Reguant and Ryan, 2016; Fowlie and Reguant, 2022), and it can make a standard preferable to a tax when emissions leak (Holland, 2012). At the border the argument reverses. A rate-based BCA is a mass-based BCA combined with a subsidy on every unit imported, so the subsidy goes to the foreign producers to which emissions leak, and the border revenue goes with it.

We proceed in two steps. A two-sector conceptual framework defines the two parts of carbon price transmission, the marginal price and the average charge, and derives qualitative insights. A general equilibrium model of the global steel industry then quantifies them. Steel accounts for roughly 7 percent of global greenhouse gas emissions and is covered by CBAM and by the US proposals (World Steel Association, 2023). It also suits the question: emissions intensities vary widely across steel plants (blast furnaces versus scrap-based electric arc furnaces) but cluster tightly in ironmaking, and iron is traded on its own, as merchant pig iron and direct reduced iron (DRI), as well as embedded in steel. The model tracks individual plants, each with its own technologies, costs, and emissions intensities, which choose input mixes and product composition and whether to make iron or buy it; finished steel and iron are traded between plants and regions. It is built from plant-level data on production, costs, technologies, and emissions, from bilateral trade flows, and from input-output and national accounts.

We compare the two BCA designs in an EU setting, where the border measure

accompanies a domestic carbon price, and a US setting, where it stands alone. For each product, the benchmark of the rate-based BCA equals the import intensity under the mass-based BCA, so imports are equally emissions intensive under both designs; trade volumes, the marginal price, emissions, welfare, and incidence are determined in equilibrium. We also compare the designs at the same global emissions reduction to determine which achieves the climate objective at lower cost.

Our main findings are as follows. First, removing the average charge weakens global abatement even when marginal incentives remain strong. For flat steel, the EU rate-based BCA offers a marginal incentive of \$81 per tCO_2 against \$88 under the mass-based design, yet leakage rises from 16.0% to 36.2%. A mass-based BCA matches the rate-based design's global emissions reduction at about one-third of its original border price in the EU setting and one-eighth in the US setting. Both designs induce resource shuffling: exporters partly meet the border requirement by redirecting cleaner output toward the importing region, without necessarily reducing emissions intensity within plants. Shuffling is strong for steel, whose intensities are dispersed, and limited for pig iron, whose intensities cluster. In the US setting, where the border measure stands alone, both designs also induce reverse leakage by shifting production home.

Second, the average charge determines who pays and who receives protection. The mass-based BCA raises \$3.57 billion in EU border revenue, partly at trading partners' expense, and protects domestic steelmakers more. The rate-based design raises steel users' prices less and transfers \$1.41 billion from dirtier to cleaner exporters, 40% within exporting regions, while collecting no net revenue. At equal import intensities, the mass-based design costs the EU less, because its border revenue more than covers its higher domestic prices, whereas the rate-based design costs the US less but also achieves less abatement. At the same global emissions reduction, however, a mass-based BCA with a lower border price costs both the importing region and the world less, setting aside administrative and legal considerations.

Third, benchmarking can leave carbon-intensive intermediates nearly uncharged, creating vertical leakage. Pig iron intensities cluster around the benchmark, so almost every exporter faces a small net charge while EU ironmakers pay the domestic carbon price. Relative to the mass-based BCA, EU pig iron imports double and their price falls by 42%; foreign merchant ironmaking accounts for 43% of the difference in foreign emissions between the two designs. The import expansion persists across the benchmark and elasticity variants we examine.

Our paper contributes to several strands of literature. It relates to work on instrument choice in environmental regulation. Policies that raise no revenue differ from emissions pricing in cost and incidence (Fullerton and Metcalf, 2001; Goulder, Hafstead and Williams III, 2016; Abrell, Rausch and Streitberger, 2019). When emissions can leak, border adjustment generally does more than output-based rebating for domestic producers (Fischer and Fox, 2012; Monjon and Quirion, 2011; Böhringer, Rosendahl and Storrøsten, 2017), although a single jurisdiction may prefer an intensity standard (Bushnell et al., 2017). Copeland, Shapiro and Taylor (2022) review these and other policies against leakage. We show that the case for a rate-based instrument reverses at the border, where its output subsidy accrues to

foreign producers. In our US setting the rate-based BCA also costs the importing region less, but because it does less, not because it limits leakage. Concurrent work by Casey, Meng and Rudik (2026) compares standalone US carbon tariffs with tariffs paired with a domestic carbon tax and, like Brunel and Levinson (2026), finds that a standalone tariff shifts production home; we draw the same contrast between our EU and US settings and vary how the border charge is computed within each.

Among studies of how to design border adjustments (Böhringer, Balistreri and Rutherford, 2012; Cosbey et al., 2019; Fowlie, Petersen and Reguant, 2021; Campolmi et al., 2025), Clausing et al. (2026) are closest to ours. They also use plant-level data for steel and quantify how a mass-based border adjustment affects competitiveness, leakage, and incidence across countries, noting that charging country averages instead of plant emissions misprices most plants. We compare mass- and rate-based designs and model trade in iron intermediates between plants. Policy proposals benchmark the charge, at the intensity of the best available technology, by exempting emissions up to a benchmark, or at domestic intensity (Ismer and Neuhoff, 2007; Pizer and Campbell, 2021; Keen, Parry and Roaf, 2022). We quantify what benchmarking gives up when the benchmark also credits cleaner exporters, so that the average charge on imports is zero in equilibrium. Keen and Kotsogiannis (2025) show that charging the difference in binding carbon prices, explicit or shadow, is efficient when technologies are the same across countries, and charging the difference in emissions intensities is efficient when binding carbon prices are equal. Both results assume fixed emissions intensities and perfectly inelastic compensated demand. In our setting emissions intensities differ widely across plants, also within countries, and respond to the border measure, as does demand for steel.

Studies find little relocation of production or emissions under early carbon pricing (Dechezleprêtre and Sato, 2017; Dechezleprêtre et al., 2022), but Coster, di Giovanni and Mejean (2025) show that firms shift imports of regulated inputs to suppliers outside the EU and, in a model with heterogeneous firms, that a CBAM-like tariff reverses this shift. We show that whether a border measure closes this channel depends on its design: a rate-based BCA leaves it open where emissions intensities cluster. Other recent trade models of carbon border adjustment also let firms differ within industries (Sogalla, 2026; Stillger, 2025); we let the border charge depend on each foreign plant's emissions intensity, to quantify resource shuffling (Bushnell, Peterman and Wolfram, 2008) and vertical leakage under the two designs.

Section I describes the steel industry and its vertical structure. Section II presents the conceptual framework, Section III the quantitative model, and Section IV the data. Section V describes the experiments, relates them to actual policies, and presents the results on emissions, who pays and who is protected, and vertical leakage through intermediates, followed by resource shuffling, reverse leakage, and sensitivity. Section VI concludes.

I. Industry Background

This section describes the features of the steel industry that determine how much of the carbon price a border measure transmits abroad and whom it protects. Steel is a natural setting because the margins emphasized in the paper—heterogeneity

in embodied emissions, vertical linkages, and trade exposure—are all economically important and measurable in this sector. Numbers in this section are from our plant data for 2019.

Three features are central. First, steel production is technologically heterogeneous, so otherwise similar products can embody very different emissions across plants and regions, which decides how much a benchmark transfers among exporters. Second, steel is produced in a vertically linked chain in which carbon-intensive intermediates such as pig iron and DRI can be traded separately from finished steel, and their emissions intensities cluster far more tightly than those of steel. Third, production technologies, input prices, and trading partners are regionally clustered, which decides where the average charge on imports falls and on whom. Together, these features explain why steel is informative about a broader class of emissions-intensive, trade-exposed (EITE) sectors in which benchmark design, trade exposure, and supply-chain boundaries jointly determine how much of the carbon price reaches foreign producers.

A. Production Routes and Embodied Emissions

Steel is produced through several routes with distinct input bundles and emissions intensities. The dominant blast furnace–basic oxygen furnace (BF-BOF) route accounts for 72% of global production. It uses iron ore and coke to produce hot metal and then refines that metal into crude steel. This route is carbon intensive, averaging 2.5 tonnes of CO_2 per tonne of crude steel, including ironmaking and electricity. It is also thermally integrated: keeping iron molten saves substantial energy, so most ironmaking remains captive to the steelmaking site and only a limited share is traded internationally as merchant pig iron. Although BF-BOF plants can add scrap to lower emissions, metallurgical and thermal constraints limit this share.

Electric arc furnace (EAF) production accounts for the remaining 28% of world output and can be much less emissions intensive when it relies on scrap and relatively clean electricity. In the US, where scrap availability is high and electricity prices have historically favored EAFs, this route accounts for 70% of production. Many EAF producers, however, do not operate on scrap alone. To meet higher-grade requirements, they often blend scrap with primary iron units such as pig iron or DRI. EAF intensities therefore range widely: the median EAF plant emits 0.3 tonnes of CO_2 per tonne of steel, while plants that rely on pig iron or DRI emit several times as much. Because scrap is limited in quantity, by the long service life of steel products, and in quality, the dispersion in embodied emissions across competing suppliers persists.

A mass-based BCA charges every exporter for its emissions; a rate-based BCA charges exporters above the benchmark and credits those below it. The wider the dispersion, the larger the transfers a rate-based BCA makes among exporters, and the more room exporters have to send cleaner output to the regulated market (resource shuffling).

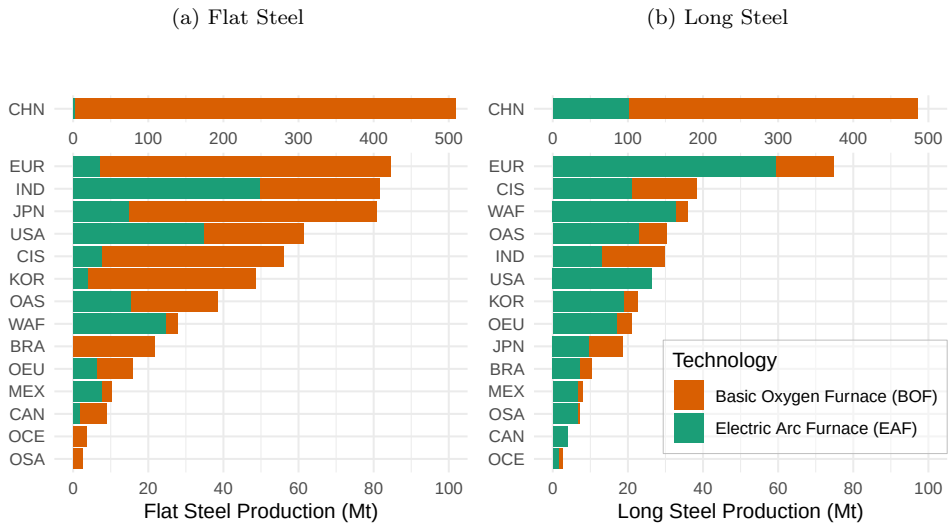


Figure 1. Steel Production by Technology and Region

Notes: Bars report production by region, with fill indicating the steelmaking route (BOF: basic oxygen furnace; EAF: electric arc furnace); production is in Mt. Region codes are defined in Section IV.A. Source: Own calculation based on CRU Steel Cost Service and World Steel Association Statistical Yearbooks.

B. Vertical Linkages

Steel has traditionally been a vertically integrated industry. Ironmaking and steelmaking are technologically linked, especially in BF-BOF production, but this linkage does not imply that all iron inputs are produced and used within the same plant or firm. In integrated BF-BOF production, iron is typically produced on site and transferred internally to the steel shop. In other parts of the industry, downstream producers purchase merchant iron inputs—most importantly pig iron and DRI—and combine them with scrap to produce steel.

Integrated BF-BOF plants have limited scope to substitute away from on-site ironmaking because molten iron charging is efficient only within the integrated facility. EAF producers, by contrast, can adjust the mix of scrap, pig iron, and DRI they use to satisfy grade requirements. This asymmetry creates a distinct upstream trade margin alongside the sourcing of finished steel. Ironmaking intensities, moreover, cluster: the intensities of the plants exporting pig iron to the EU vary by about 11.7% around their mean, far less than those of steel (Table C1).

A border measure that benchmarks a homogeneous intermediate can leave imports of it almost uncharged while domestic producers pay the carbon price. Downstream steel users then substitute toward imported intermediates, and protection shifts away from domestic upstream producers.

C. Geography of Costs, Emissions, and Trade

Figure 1 illustrates the geographic clustering of these technologies. With the exception of China and India, long products are mainly produced by EAF mills. Flat

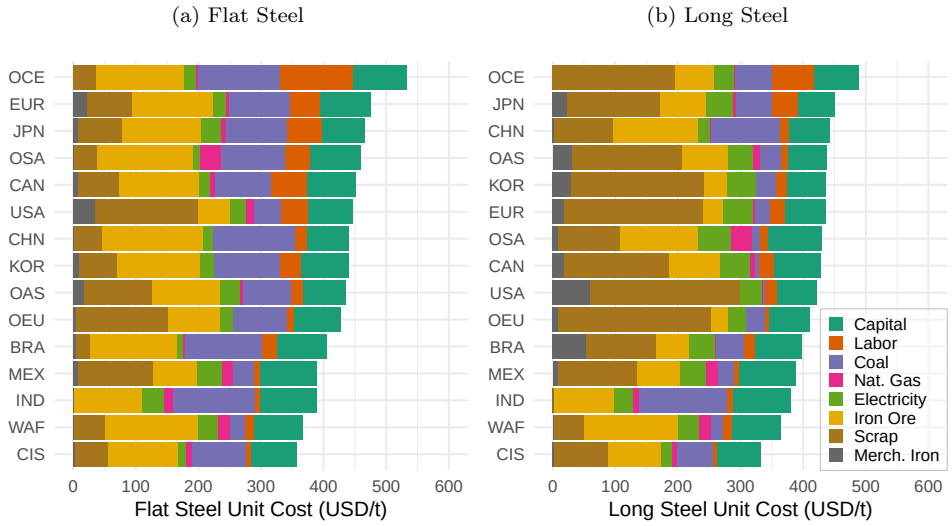


Figure 2. Average Steel Production Cost Structure by Region

Notes: Bars decompose unit costs (USD/t) by input category for each region. Region codes are defined in Section IV.A. Source: Own calculation based on CRU Steel Cost Service.

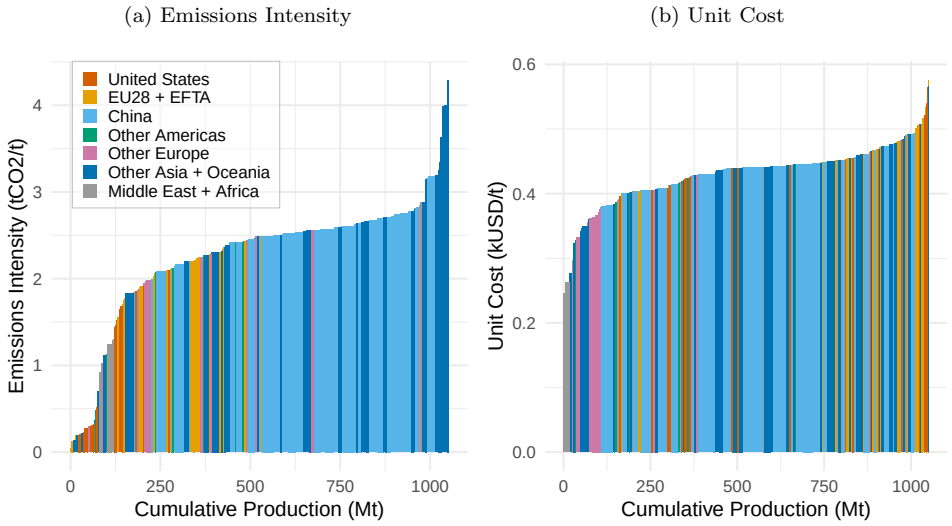


Figure 3. Cost and Emissions Intensity Curve for Flat Steel

Notes: Each panel orders flat-steel plants from lowest to highest value; bar width equals plant production, so the horizontal axis is cumulative production (Mt). Bars are colored by region group (see legend). Panel (a) reports emissions intensity (tCO_2/t) and panel (b) unit cost (thousand USD/t). Source: Own calculation based on CRU Steel Cost Service.

products are predominantly produced by BF-BOF plants, with exceptions including the US, Mexico, India, and Western Asia and Africa, where EAF mills produce more

than half.

Figure 2 shows that this technological geography carries over into costs. In the US, a relatively large share of production costs comes from third-party scrap purchases in both long and flat products, reflecting extensive EAF use and a developed recycling system. Regions dominated by BF-BOF production instead devote more of their cost base to iron ore and reductant inputs such as coking coal. More important for our purposes, Figure 3 extends this to emissions intensity: in flat steel, North American and European plants have higher production costs but lower CO_2 intensities, owing to cleaner technologies, whereas plants in Asia combine lower costs with higher emissions.

When cleaner producers are also higher cost, the average charge falls mostly on low-cost, dirtier imports: a mass-based BCA charges the average tonne imported and therefore suppresses demand more strongly, but it also raises downstream prices more sharply. A benchmarked, rate-based design preserves more low-cost trade and softens downstream price increases, but it removes the average charge on imports and so weakens the scale margin of abatement.

Trade exposure is similar in the two importing regions of our experiments, but its sources differ. The EU and the US each import about a fifth of the steel they consume; the EU imports mainly from Turkey and Other Europe, the CIS region, and South Korea, and the US from Canada, the EU, and South Korea (Table B2). The same border design therefore meets different exporters in the two settings.

Taken together, these industry features motivate the modeling choices we make in Section III. Plant-level heterogeneity is needed to capture how alternative border designs interact with wide dispersion in embodied emissions. Import demand defined over individual foreign plants is needed so that each exporter's charge reaches its own price. Explicit merchant iron markets are needed to quantify substitution toward upstream intermediates and the resulting leakage channels. A general-equilibrium framework is needed to track who pays among upstream producers, downstream users, and foreign suppliers.

II. Conceptual Framework

This section develops a two-sector framework that carries the logic of a domestic intensity standard, an emissions tax combined with an output subsidy, to the border, where the subsidy is paid to foreign producers. The framework delivers the two objects used throughout the quantitative analysis: how much of the carbon price is transmitted to foreign producers, at the margin and on average, and whom the border measure protects along the supply chain, upstream producers or downstream users, and who pays for it. It derives one result for each of our three contributions. Appendix A contains the proofs and two supporting results.

A. Setup

To isolate the margins emphasized in the introduction, we consider a vertically linked economy with two sectors: an upstream sector and a domestic downstream sector. The upstream sector spans two regions, home (h) and foreign (f). We

distinguish between the domestic carbon price $\tau_h \geq 0$ faced by home producers and the border price $\tau^B > 0$ applied to imports from the foreign region. This setup nests both the EU setting ($\tau_h > 0$), in which the border measure accompanies a domestic carbon price, and the US setting ($\tau_h = 0$), in which it stands alone.

UPSTREAM SECTOR.—Each regional upstream sector consists of clean (c) and dirty (d) plants with emissions intensities $\epsilon_c < \epsilon_d$ and base unit production costs $c_c > c_d$. A plant cannot change its intensity, so a region's emissions intensity changes only through the share of its output that comes from clean plants. In the quantitative model, plants can also lower their intensity by substituting among inputs, but this technique effect is small and similar under the two designs (Section V.E). The sectoral unit cost of production $c(s_{c,i})$ depends on the regional clean share $s_{c,i} \in [0, 1]$ and is strictly increasing and convex ($c' > 0, c'' > 0$), reflecting the higher base cost of clean plants and the rising cost of shifting output toward them as their capacity is used up.

Home upstream producers supply volume $q_{h,h}$ exclusively to the regulated domestic market, while foreign upstream producers serve both the regulated market ($q_{f,h}$) and an unregulated foreign domestic market ($q_{f,f}$). We consider inelastic foreign domestic demand $q_{f,f}$ to isolate the policy-driven trade-offs between the clean share and trade scale within the regulated market. Let $\varphi_{i,j} = q_{i,j}/(q_{i,h} + q_{i,f})$ denote the share of destination $j \in \{h, f\}$ in the total production of region i, where $\varphi_{h,h} = 1$ and $\varphi_{h,f} = 0$. The border charge $t(\cdot)$ applies only to the volume sold in the regulated market ($j = h$) based on its specific intensity $\epsilon_{i,h} = s_{c,i,h}\epsilon_c + (1 - s_{c,i,h})\epsilon_d$. The representative producer in region i chooses the clean shares for each market $(s_{c,i,h}, s_{c,i,f})$ to minimize the unit cost of the variety sold in the regulated market:

$$\min_{s_{c,i,h}, s_{c,i,f}} \{c(s_{c,i}) + t(\epsilon_{i,h})\} \tag{1}$$

where $s_{c,i} = \sum_j \varphi_{i,j} s_{c,i,j}$ is the regional clean share. The instrument $t(\epsilon)$ is determined by the policy design. Under a mass-based design, $t^M(\epsilon) = \tau^B \epsilon$. Under a rate-based design, $t^R(\epsilon) = \nu(\epsilon - \bar{\psi})$, where $\bar{\psi}$ denotes the intensity benchmark and $\nu \geq 0$ is the credit price, which is determined in equilibrium: imports may be no more emissions intensive than the benchmark, $\epsilon_{f,h} \leq \bar{\psi}$, and $\nu = 0$ if this constraint does not bind. In equilibrium, the price $p_{i,h}$ of the regional variety in the regulated market is $p_{i,h} = c(s_{c,i}) + t(\epsilon_{i,h})$.

DOWNSTREAM SECTOR.—Domestic downstream firms combine domestic ($q_{h,h}$) and foreign ($q_{f,h}$) upstream varieties to produce a composite intermediate Q. The aggregation follows a linearly homogeneous and concave function $Q = F(q_{h,h}, q_{f,h})$. Let $P(p_{h,h}, p_{f,h})$ denote the associated dual price index, representing the minimum cost of producing one unit of Q. This index is strictly increasing ($\partial P/\partial p_{i,h} > 0$) and concave ($\partial^2 P/\partial p_{i,h}^2 \leq 0$) in input prices. Downstream firms choose Q to maximize producer surplus $\Pi_d = \max_Q\{G(Q) - P(p_{h,h}, p_{f,h})Q\}$, where $G(Q)$ represents the value-added of the upstream input in downstream production. By Shephard's Lemma, the demand for each variety is given by $q_{i,h} = Q \cdot \partial P/\partial p_{i,h}$. The concavity of P ensures that demand for each variety is strictly decreasing in its own price

$p_{i,h}$. This downstream block is intentionally parsimonious: its role is to map border-induced changes in upstream prices into trade volumes and into the incidence on upstream producers and downstream users. This general structure can be specialized to a constant elasticity of substitution (CES) functional form, consistent with standard Armington trade assumptions.

The two designs differ in how they charge imports. The next lemma separates the two parts of carbon price transmission.

LEMMA 1 (Marginal Price and Average Charge): *Consider imports from the foreign region with intensity $\epsilon_{f,h}$.*

(i) Marginal price: a unit increase in intensity raises the charge per unit by τ^B under the mass-based design and by ν under the rate-based design.

(ii) Average charge: the charge per unit imported is $\tau^B \epsilon_{f,h} > 0$ under the mass-based design and $\nu(\epsilon_{f,h} - \bar{\psi}) = 0$ under the rate-based design, under which the charge on dirty plants' output, $\nu(\epsilon_d - \bar{\psi})$ per unit, is credited to clean plants' output.

(iii) Decomposition: let the benchmark equal the intensity of imports under the mass-based design, $\bar{\psi} = \epsilon^M_{f,h}$, and bind. Then imports are equally emissions intensive under both designs, and the rate-based design charges $\tau^B \bar{\psi}$ less per unit imported, whatever the credit price: it equals the mass-based design combined with a subsidy of $\tau^B \bar{\psi}$ on the average unit imported.

PROOF: See Appendix A.1.

Part (iii) carries the decomposition of an intensity standard into an emissions tax and an output subsidy (Fischer, 2001; Holland, Hughes and Knittel, 2009) to the border, where its implication reverses. Within one economy, the subsidy goes to regulated producers and limits leakage (Holland, 2012); at the border, it goes to the foreign producers to which emissions leak. Part (ii) holds at the equilibrium intensity of imports, not at the intensity before the policy. If the benchmark lies below the pre-policy intensity of imports, the rate-based design still raises the import price, because meeting the benchmark requires a higher and costlier clean share. The propositions below compare the designs at the same intensity of imports, as in part (iii) and in the experiments of Section V. Imports are then equally clean under both designs, so the comparison isolates the average charge; the credit price is whatever the benchmark requires.

B. Emissions

PROPOSITION 1 (Emissions): *Compare the two designs at the same intensity of imports, $\bar{\psi} = \epsilon^M_{f,h}$.*

(i) The price of imports is lower under the rate-based design, which removes the average charge $\tau^B \bar{\psi}$, so imports and the composite Q are larger.

(ii) Global emissions are higher under the rate-based design if the additional imports embody more emissions than the home output they displace.

(iii) Let $\tau_h = \tau^B$. The mass-based design achieves the global emissions of the rate-based design at lower cost, unless demand for imports is perfectly inelastic: it prices every margin of abatement at the same rate, whereas the rate-based design leaves the volume of imports unpriced.

PROOF: See Appendix A.2.

The marginal price does the same work under both designs: it raises the clean share of exports to the same level, partly through more output from clean plants and partly through reshuffling (Appendix A.5). What the rate-based design gives up is the scale margin: without the average charge, imports are cheaper and larger. Part (iii) is the sense in which the average charge is worth a large part of the price, and it motivates the equivalent price used in Section V. In the US setting, home output bears no carbon price; Appendix A.5 shows how reshoring of home production then dilutes abatement.

C. Who Pays and Who Is Protected

PROPOSITION 2 (Who Pays and Who Is Protected): *Compare the two designs at the same intensity of imports, $\bar{\psi} = \epsilon^M_{f,h}$.*

(i) Border revenue: the home government collects $\tau^B \bar{\psi} q_{f,h}$ under the mass-based design and nothing under the rate-based design, under which the charges on dirty plants' output are credited to clean plants' output.

(ii) Upstream: the import price is higher under the mass-based design, so home producers' share of the regulated market is larger.

(iii) Downstream: the rate-based design lowers the input price index $P(p_{h,h}, p_{f,h})$, thereby increasing the surplus of downstream firms: $\Pi_d(P^R) > \Pi_d(P^M)$.

PROOF: See Appendix A.3.

The average charge therefore decides who pays. Under the mass-based design it becomes border revenue and protects home producers at the expense of downstream users; under the rate-based design it stays among foreign producers, moving from dirty to clean plants. In this framework foreign supply is perfectly elastic in volume, so the charge is passed on to home buyers in full. Home welfare adds border revenue, producer surplus, and downstream surplus, which move in different directions, so the framework does not rank the designs.

D. Intermediates and Vertical Leakage

The upstream good in this framework is an intermediate, and the logic applies with particular force where its producers' intensities cluster, as for pig iron.

PROPOSITION 3 (Intermediates and Vertical Leakage): *Suppose the intensities of clean and dirty plants lie within δ of the benchmark, $|\epsilon_j - \bar{\psi}| \leq \delta$ for $j \in \{c, d\}$, and compare the two designs at the same intensity of imports in the EU setting ($\tau_h > 0$).*

(i) Under the rate-based design, every imported unit is charged or credited at most $\nu\delta$, while home producers pay $\tau_h\epsilon_{h,h}$ per unit; under the mass-based design, imports pay $\tau^B\bar{\psi}$ per unit on average.

(ii) If production costs are similar across regions and $\nu\delta$ is small, the rate-based design makes imported intermediates cheaper than home intermediates by close to the full domestic carbon cost $\tau_h\epsilon_{h,h}$, and downstream firms substitute toward imported intermediates.

PROOF: See Appendix A.4.

The mass-based design charges imports in full, so that with $\tau^B = \tau_h$ home and imported intermediates of similar intensity bear similar carbon costs. Under the rate-based design, charges and credits are small when intensities cluster around the benchmark and the credit price is moderate. A benchmark well below the intensity of most plants can be reached only through large shifts among similar plants, and the credit price rises steeply as the benchmark approaches the intensity of the cleanest plants. We call the substitution in part (ii) vertical leakage: emissions move abroad through the upstream stage as domestic downstream firms buy imported intermediates.

III. Quantitative Model

This section translates the industry features in Section I and the mechanisms in Section II into a quantitative-empirical framework. For our question, the model must capture four features: emissions intensities at the plant level, on which the marginal price acts; import prices specific to each foreign plant, through which the average charge reaches prices; merchant iron, which can replace domestic ironmaking; and general equilibrium, which determines who pays.

OVERVIEW.—We develop a multi-region general equilibrium model of the global steel supply chain (see Figure 4). Regional households supply labor, capital, and raw materials and demand final consumption. A macroeconomic production block bundles steel, energy, and other intermediates into final goods. Its distinctive feature is a granular industrial layer: individual iron and steel plants operate different technologies, face plant-specific costs and emissions intensities, and supply differentiated products to domestic and foreign markets. The model also captures the industry's vertical structure by allowing iron to be produced on-site and fed into steelmaking or purchased through the merchant iron market.

Climate-trade policy enters through regional carbon pricing on production inputs and BCAs on the embodied carbon content of imports, implemented either on a mass basis or relative to an intensity benchmark. Revenues are rebated to households, and prices and quantities are determined jointly across factor, product, and intermediate-input markets and international trade.

A. The Industrial Layer: Steel and Ironmaking

PLANTS, TECHNOLOGIES, AND VARIETIES.—The iron and steel sector is composed of a set of individual production facilities $i \in \mathcal{I}$. Each plant i operates a portfolio of

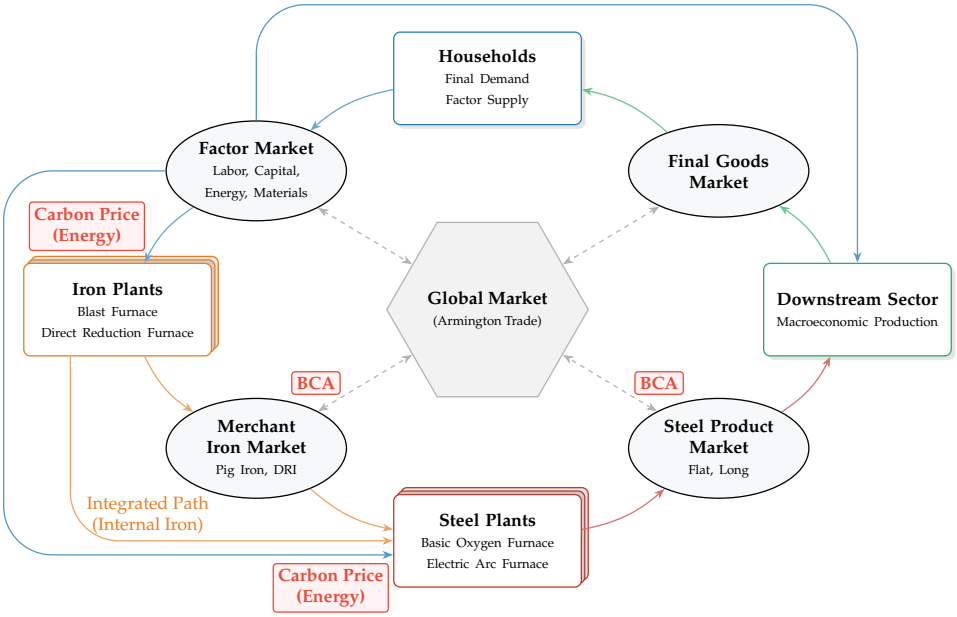


Figure 4. Model Structure

available technologies $a \in \mathcal{A}_i \subseteq \mathcal{A}$. We explicitly partition the metallurgical supply chain into ironmaking and steelmaking stages: $\mathcal{A} = \mathcal{A}_I \cup \mathcal{A}_S$, where $\mathcal{A}_I = \{\text{BF}, \text{DRF}\}$ and $\mathcal{A}_S = \{\text{BOF}, \text{EAF}\}$. The corresponding outputs are partitioned into intermediate iron varieties $v \in \mathcal{V}_I = \{\text{Pig Iron}, \text{DRI}\}$ and finished steel varieties $v \in \mathcal{V}_S = \{\text{Flat}, \text{Long}\}$. Plants act as price-takers under perfect competition, with their output levels determined by site-specific marginal costs. This separation between upstream intermediates and finished steel is what allows the quantitative model to track both reshuffling in final products and leakage through merchant iron markets.

PRIMARY PRODUCTION AND UNIT COSTS.—Production at the plant level is governed by a multi-tier nested CES cost structure. This formulation captures the hard-to-abate character of steel production by enforcing the physical limits of metallurgy while allowing for price-induced substitution across energy, capital, labor, and materials. At the plant level i using technology $a \in \mathcal{A}$, producers combine a material composite with a capital-labor-energy (KLE) bundle to produce semi-finished iron or steel. Under the assumption of perfect competition and constant returns to scale, the producer price of semi-finished products is equal to its unit cost $c_{i,a}(\boldsymbol{\omega}_{i,a,f}, p_{i,a,e})$. The top-level unit cost is defined as:

$$c_{i,a} = \left[\theta_{i,a}^{\text{Mat}} (p_{i,a}^{\text{Mat}})^{1-\sigma_a} + (1 - \theta_{i,a}^{\text{Mat}})(p_{i,a}^{\text{KLE}})^{1-\sigma_a}\right]^{\frac{1}{1-\sigma_a}} \tag{2}$$

where $p_{i,a}^{\text{Mat}}$ is the price of the material composite, $p_{i,a}^{\text{KLE}}$ is the cost of the KLE bundle, $\theta_{i,a}^{\text{Mat}}$ is the benchmark material value share, and σ_a is the top-level elasticity of substitution, calibrated to a near-Leontief value to reflect metallurgical mass balance.

The cost of the KLE bundle, $p_{i,a}^{\text{KLE}}$, aggregates value-added and energy:

$$p_{i,a}^{\text{KLE}} = \left[\theta_{i,a}^{\text{VA}}(p_{i,a}^{\text{VA}})^{1-\sigma_a^{\text{KLE}}} + (1-\theta_{i,a}^{\text{VA}})(p_{i,a}^{\text{Ene}})^{1-\sigma_a^{\text{KLE}}}\right]^{\frac{1}{1-\sigma_a^{\text{KLE}}}} \tag{3}$$

where $p_{i,a}^{\text{VA}}$ is the value-added cost, $p_{i,a}^{\text{Ene}}$ is the composite energy price, $\theta_{i,a}^{\text{VA}}$ is the benchmark value-added share, and σ_a^{KLE} is the substitution elasticity between energy and value-added. The value-added cost $p_{i,a}^{\text{VA}}$ is a composite of plant-specific factor prices $\boldsymbol{\omega}_{i,a,f}$ (e.g., labor and capital), while the energy bundle cost $p_{i,a}^{\text{Ene}}$ aggregates energy carriers $e \in \mathcal{E}$, where $\mathcal{E} = \{\text{coal}, \text{natural gas}, \text{electricity}\}$.[1]

The user cost of each energy carrier $p_{i,a,e}$ reflects the regional carbon price τ_r:

$$p_{i,a,e} = \omega_{i,a,e} + \tau_r \epsilon_{i,a,e}, \tag{6}$$

where $\omega_{i,a,e}$ is the plant-specific pre-tax cost of energy carrier e (governed by the mobility frontier defined in 4), τ_r is the regional carbon price, and $\epsilon_{i,a,e}$ is the carbon intensity per unit of energy (measured in tCO_2 per physical unit of energy carrier e). Note that we distinguish between the plant's emissions intensity per energy source, $\epsilon_{i,a,e}$, and its product-level intensity, $\epsilon_{i,a,v}$ (measured in tCO_2 per tonne of output), which accounts for the full metallurgical process including upstream iron production. That distinction is central because BCAs apply to embodied product emissions, not directly to individual energy inputs. The limited substitutability within these nests ($\sigma_a^{\text{KLE}}, \sigma_a^{\text{Ene}}$) formally represents the sector's hard-to-abate nature, as plants cannot easily switch away from carbon-intensive fuels in response to τ_r.

TRANSFORMATION INTO INTERMEDIATE AND FINAL VARIETIES.—Semi-finished products are transformed into variety-specific outputs $v \in \mathcal{V}$ according to a CET frontier. For ironmaking technologies $a \in \mathcal{A}_I$, this yields intermediate iron varieties ($v \in \mathcal{V}_I$), while steelmaking technologies $a \in \mathcal{A}_S$ produce finished steel varieties ($v \in \mathcal{V}_S$). The supply of a specific variety v from plant i is given by:

$$q_{i,a,v} = \theta_{i,a,v} Q_{i,a} \left(\frac{p_{i,a,v}}{c_{i,a}}\right)^{\eta_a^V} \tag{7}$$

[1] We allow for idiosyncratic price dispersion at the plant level to reflect local market frictions. Regional aggregate supplies—whether primary endowments or Armington composites—are transformed into plant-specific inputs ($\omega_{i,a,f}$) via a constant elasticity of transformation (CET) frontier:

$$\omega_{i,a,f} = \Omega_{r,f}\left[\theta_{i,a,f}^F \left(\frac{q_{i,a,f}}{Q_{r,f}}\right)\right]^{\frac{1}{\eta_f^F}}, \tag{4}$$

where $q_{i,a,f}$ is the quantity of input f allocated to plant i and η_f^F is the transformation elasticity governing mobility. This captures the sluggishness in reallocating specialized resources, including localized constraints in logistics or contracting for energy and materials, between specific industrial sites.

Input demand can be derived by differentiating the unit cost function with respect to input prices according to Shephard's Lemma:

$$q_{i,a,f} = Q_{i,a} \frac{\partial c_{i,a}}{\partial \omega_{i,a,f}}, \tag{5}$$

where $Q_{i,a}$ is the total activity level of the plant.

where $q_{i,a,v}$ is the quantity of variety v produced, $Q_{i,a}$ is the total activity level of the plant, $p_{i,a,v}$ is the variety price, η_a^V is the transformation elasticity, and $\theta_{i,a,v}$ is the benchmark volume share of variety v in technology a's output.

VERTICAL SUPPLY CHAINS AND IRON SOURCING.—A defining feature of the model is the vertical linkage between the iron and steel stages. Because carbon emissions are primarily concentrated in the ironmaking stage, steelmaking technologies ($a \in \mathcal{A}_S$) face a strategic supply chain choice: produce iron intermediate inputs $v \in \mathcal{V}_I$ internally (onsite integration) or purchase them from the regional merchant market. The effective user cost $p_{i,a,v}^{\text{Iron}}$ for a steelmaker is a CES composite:

$$p_{i,a,v}^{\text{Iron}} = \left[\theta_{i,a,v}^{\text{Iron}}(p_{i,a',v})^{1-\sigma_v^{\text{Iron}}} + (1-\theta_{i,a,v}^{\text{Iron}})(P_{r,v})^{1-\sigma_v^{\text{Iron}}}\right]^{\frac{1}{1-\sigma_v^{\text{Iron}}}} \tag{8}$$

where $p_{i,a',v}$ is the cost of iron produced onsite using technology $a' \in \mathcal{A}_I$, $P_{r,v}$ is the regional price of the Armington iron composite, $\theta_{i,a,v}^{\text{Iron}}$ is the plant-specific benchmark onsite share, and σ_v^{Iron} is the substitution elasticity between onsite and merchant iron.

B. Linking Plants to Output and Input Markets

This block links the industrial layer to regional goods and factor markets and to downstream demand.

TRADE IN STEEL AND RAW MATERIALS.—In each region r, the total supply of steel varieties and tradable raw materials (factors $f \in \mathcal{F}_{\text{tradable}}$) is formed by aggregating domestic and imported quantities through a nested CES (Armington) structure. For each commodity $j \in \mathcal{V} \cup \mathcal{F}_{\text{tradable}}$, the regional price index $P_{r,j}$ is:

$$P_{r,j} = \left[\theta_{r,j}^A (P_{r,j}^D)^{1-\sigma_j^A} + (1-\theta_{r,j}^A)(P_{r,j}^M)^{1-\sigma_j^A}\right]^{\frac{1}{1-\sigma_j^A}}, \tag{9}$$

where $P_{r,j}^D$ and $P_{r,j}^M$ are the prices of the domestic and imported composites, respectively. The domestic composite price $P_{r,j}^D$ aggregates local supply from plants $i \in \mathcal{I}_r$ for steel, or local resource extraction for raw materials. The imported bundle price $P_{r,j}^M$ similarly aggregates imports across foreign origins using the substitution elasticity σ_j^M; for raw materials, the origins are foreign regions $s \neq r$:

$$P_{r,j}^M = \left[\sum_{s \neq r} \theta_{s,r,j}^M (p_{s,r,j})^{1-\sigma_j^M}\right]^{\frac{1}{1-\sigma_j^M}}, \tag{10}$$

where $p_{s,r,j}$ is the price of commodity j exported from region s to region r. For steel and iron, the origins are individual foreign plants:

$$P_{r,v}^M = \left[\sum_{i \notin \mathcal{I}_r} \sum_{a \in \mathcal{A}} \theta_{i,r,a,v}^M (p_{i,r,a,v}^x)^{1-\sigma_v^M}\right]^{\frac{1}{1-\sigma_v^M}}, \tag{11}$$

where $p^x_{i,r,a,v}$ is the price at which plant i sells variety v in region r, including any border charge (Section III.C), so each plant's charge enters its own price. This Armington structure ensures that the model captures the idiosyncratic trade patterns and cross-border price transmission for both industrial goods and upstream inputs, and governs how strongly border charges pass through into destination-market prices and trade shares.[2]

MACRO-PRODUCTION.—The aggregate macroeconomic sector (Mac) produces the intermediate macro-good $Q_{r,Mac}$ by bundling primary factors, macro-intermediates, and Armington steel varieties $v \in \mathcal{V}_S$. The unit cost $P_{r,Mac}$ is the dual of the nested CES production function:

$$P_{r,Mac} = \mathcal{C}_{r,Mac}(\Omega_{r,f}, P_{r,v}, \dots), \tag{13}$$

and the resulting demand for each steel variety v is $Q_{r,Mac,v} = Q_{r,Mac}(\partial P_{r,Mac}/\partial P_{r,v})$. Finally, the consumption bundle $Q_{r,C}$ is produced using the macro-good as the primary input, at price $P_{r,C}$. This block is what maps changes in steel prices into downstream user prices, aggregate demand, and ultimately welfare.

MARKETS FOR STEEL VARIETIES AND MATERIAL INPUTS.—For each steel variety and tradable raw material $j \in \mathcal{V} \cup \mathcal{F}_{\text{tradable}}$, the total quantity supplied by region r must equal the sum of domestic demand and aggregate export demand:

$$Q^{\text{sup}}_{r,j} = q^D_{r,r,j} + \sum_{s \neq r} q^M_{r,s,j}, \tag{14}$$

where $Q^{\text{sup}}_{r,j}$ is the total regional production (summed across plants for steel, or local resource extraction for materials), $q^D_{r,r,j}$ is domestic demand, and $q^M_{r,s,j}$ denotes aggregate bilateral exports from region r to destination s. The global equilibrium requires that these bilateral exports strictly equal the sum of import demands from the destination region. For steel varieties v, this bilateral trade identity is:

$$q^M_{r,s,v} = \sum_{i \in \mathcal{I}_r} \sum_{a \in \mathcal{A}} q_{i,s,a,v}, \tag{15}$$

which formally maps the plant-to-region derived demand from (12) into the regional export flow.

MARKET FOR MACRO GOOD.—The regional market for the macroeconomic good Mac clears when the total output $Q_{r,Mac}$ equals the sum of intermediate demand from the

[2]The demand for a specific plant-level variety of finished steel $q_{i,r,a,v}$ in destination region r is derived from the Armington composites via Shephard's Lemma. For a given total regional demand $Q_{r,v}$, the quantity sourced from plant i in region s is:

$$q_{i,r,a,v} = Q_{r,v} \frac{\partial P_{r,v}}{\partial p^x_{i,r,a,v}}. \tag{12}$$

steel sector and final consumption demand from the household:

$$Q_{r,Mac} = \sum_{i,a} q_{i,a,Mac} + Q^{C}_{r,Mac}, \tag{16}$$

where $Q^{C}_{r,Mac}$ is the quantity of the macro-good used to produce the final consumption bundle $Q_{r,C}$.

PRIMARY FACTOR MARKETS.—Each region r is endowed with a set of primary factors $f \in \mathcal{F}$, including labor, capital, and raw materials (e.g., iron ore, coking coal, and steel scrap). Labor and capital are supplied perfectly inelastically, while raw materials are supplied price-elastically. The factor market for f is in equilibrium if supply $Q_{r,f}$ equals the aggregate demand from all production activities:

$$Q_{r,f} = \sum_{i,a} q_{i,a,f} + q_{r,Mac,f}, \tag{17}$$

where $q_{i,a,f}$ is the plant-level factor demand and $q_{r,Mac,f}$ is the demand from the macro-sector.

C. Climate-Trade Policies

This block describes how the climate-trade policy instruments are represented in the quantitative model. Domestic carbon pricing raises the cost of carbon-intensive production at source, while the BCA prices embodied emissions at the trade interface either on a mass basis or relative to a benchmark.

REGIONAL CARBON PRICING ON ENERGY.—The regional carbon price τ_r enters production costs through the user cost of energy (6). Total regional emissions from the steel sector $Q_{r,CO2}$ are:

$$Q_{r,CO2} = \sum_{i \in \mathcal{I}_r} \sum_{a \in \mathcal{A}} \sum_{e \in \mathcal{E}} \epsilon_{i,a,e} q_{i,a,e}, \tag{18}$$

where $\epsilon_{i,a,e}$ is the carbon intensity per unit of energy carrier e.

BORDER CARBON ADJUSTMENTS ON EMBODIED CARBON.—While regional pricing applies to inputs, BCAs are applied to the total carbon embodied in imported varieties v, finished steel and iron alike. The embodied emissions intensity $\epsilon_{i,a,v}$ accounts for direct process emissions and energy-related emissions across the metallurgical supply chain. Under a mass-based BCA, the destination region r levies a charge on imports from plant i in region s at the border price τ^{B}_{r}:

$$p^{x}_{i,r,a,v} = p_{i,a,v} + \tau^{B}_{r} \epsilon_{i,a,v}. \tag{19}$$

The border price equals the regional carbon price τ_r in the EU setting and a reference price in the US setting (Section V.A). Under a rate-based BCA, the destination region sets an intensity target $\bar{\psi}_{r,v}$. Exporters must surrender permits covering their actual

emissions $\epsilon_{i,a,v}$ while receiving credits based on the target $\bar{\psi}_{r,v}$. The export price is:

$$p^x_{i,r,a,v} = p_{i,a,v} + \nu_{r,v}(\epsilon_{i,a,v} - \bar{\psi}_{r,v}), \tag{20}$$

where $\nu_{r,v}$ is the credit price, the marginal price on intensity under the rate-based design. It is determined endogenously by the permit market: imports from all regions $s \neq r$ may be no more emissions intensive than the target,

$$\sum_{s \neq r} \sum_{i \in \mathcal{I}_s} \sum_{a \in \mathcal{A}} (\epsilon_{i,a,v} - \bar{\psi}_{r,v}) q_{i,r,a,v} \leq 0, \tag{21}$$

and $\nu_{r,v} \geq 0$ is strictly positive only if this constraint binds. When $\nu_{r,v} > 0$, the charges on imports above the target equal the credits on imports below it, so the rate-based BCA collects no net revenue: it keeps the marginal price on intensity but removes the average charge (Lemma 1). The discounted mass-based BCA applies (19) at a lower border price, the *equivalent price*, chosen so that global emissions equal those under the rate-based BCA; its ratio to the border price of the mass-based BCA is the *equivalent price ratio* λ. All policy-generated revenues, including carbon tax proceeds and border charges, are returned lump-sum to households in the same region.

D. Aggregate Resource Constraints

The model is closed by the budget constraint of the representative household in each region. Households receive income I_r from factor returns, the benchmark trade imbalance $\bar{\Delta}_r$, and revenues from climate-trade policy:

$$I_r = \sum_{f \in \mathcal{F}} \omega_{r,f} E_{r,f} + \bar{\Delta}_r + TR_r, \tag{22}$$

where $E_{r,f}$ is the resource endowment and TR_r represents total transfers from carbon pricing and border adjustments. TR_r denotes total regional transfers defined as:

$$TR_r = \tau_r Q_{r,CO2} + \sum_{v \in \mathcal{V}} \sum_{s \neq r} \sum_{i \in \mathcal{I}_s} \sum_{a \in \mathcal{A}} (p^x_{i,r,a,v} - p_{i,a,v}) q_{i,r,a,v}, \tag{23}$$

where the first term represents domestic carbon pricing revenue and the second term represents the net proceeds (or "feebate" balance) from BCAs on all imported varieties.

Households spend this income on the aggregate final consumption bundle $Q_{r,C}$:

$$P_{r,C}\, Q_{r,C} = I_r, \tag{24}$$

where the consumer price index $P_{r,C}$ is endogenously determined by the underlying macro-production technology.

E. Equilibrium

For a given set of climate policy instruments—characterized by regional carbon prices τ_r, border prices τ_r^B, intensity targets $\bar{\psi}_{r,v}$, and permit-clearing rules—a competitive equilibrium is defined as a vector of prices, quantities, and regional incomes,

$$\mathcal{P} = \{p_{i,a,v}, P_{r,j}, \Omega_{r,f}, \omega_{r,f}, P_{r,Mac}, P_{r,C}, \nu_{r,v}\},$$
$$\mathcal{Q} = \{q_{i,r,a,v}, Q_{r,j}, Q_{r,f}, Q_{r,Mac}, Q_{r,C}\}, \qquad \mathcal{I} = \{I_r\},$$

such that consumers and firms optimize and all markets clear. Households maximize utility subject to the budget constraint (22), which determines demand for the final consumption bundle $Q_{r,C}$. Firms take prices as given and choose inputs and outputs to maximize profits, equivalently minimize costs under constant returns to scale. At the plant level, this yields unit costs $c_{i,a}$ and input demands governed by (2)–(5); at the aggregate level, the macro sector minimizes the cost of producing $Q_{r,Mac}$ according to (13). Under perfect competition, these firm optimization conditions imply zero profits and prices equal to corresponding unit costs, including variety prices $p_{i,a,v}$ and Armington price indices $P_{r,j}$. Market-clearing identities hold globally for steel varieties and material inputs (14), with bilateral trade satisfying (15), and regionally for the macro good (16) and factor markets (17). In regimes with rate-based BCAs, the credit price $\nu_{r,v}$ clears the intensity benchmark according to (21).

IV. Data and Calibration

This section describes the empirical foundation of the quantitative model. We first summarize the data sources used to measure plant-level heterogeneity, trade linkages, and the broader macroeconomic environment. We then explain how these sources are combined to construct the benchmark equilibrium that anchors the counterfactual analysis, and finally summarize the calibration of the structural parameters that govern behavioral responses.

A. Data Sources

The empirical foundation of the model combines plant-level industrial data, international trade statistics, and national accounts.

INDUSTRIAL MICRO-DATA.—Data for the industrial layer are drawn primarily from the CRU Steel Cost Service, which reports production volumes, technology portfolios, unit costs, and detailed energy and material input coefficients for individual steel mills. After cleaning, our sample contains 287 plants in 48 countries. We use the Global Steel Plant Tracker (GSPT) to validate the coverage of the CRU sample (Figure B1), and worldsteel statistics to validate regional production totals. In the cleaned sample, CRU covers 55.9% of world production, 70.3% outside China, and 98.0% of the BF-BOF segment outside China (CRU Group, 2019; Global Energy Monitor, 2023; World Steel Association, 2022).

TRADE AND MARGINS.—Bilateral trade flows for finished steel and merchant iron are sourced from UN Comtrade at the 6-digit level of the Harmonized System (HS).

Trade in primary raw materials is assembled from UN Comtrade for iron ore and scrap, the U.S. Energy Information Administration (EIA) for metallurgical coal, and the BP Statistical Review for natural gas. To discipline exporter participation and trade margins, we use OECD sources on firm counts, exporting behavior, and international transport costs (United Nations Statistics Division, 2024; U.S. Energy Information Administration, 2024; BP, 2022; OECD, 2021, 2024*b*,*c*,*a*).

MACROECONOMIC FRAMEWORK.—The broader macroeconomic environment is anchored to version 11 of the Global Trade Analysis Project (GTAP) database (Aguiar et al., 2022), which provides the national accounts and trade structure used to embed the steel sector in general equilibrium.[3]

B. Constructing the Benchmark Equilibrium

We combine these sources to construct the benchmark equilibrium. Throughout the remainder of this section, benchmark data objects corresponding to endogenous model variables are denoted with an overbar. The benchmark allocation is therefore summarized by $\{\overline{Q}_{i,a}, \overline{q}_{i,a,f}, \overline{q}_{i,s,a,v}, \overline{\epsilon}_{i,a,v}\}$, where the overbar indicates the factual reference equilibrium. The construction proceeds in four steps.

SAMPLE SELECTION AND VALIDATION.—We begin by cleaning the raw plant records, dropping facilities with incomplete or inconsistent operational accounts, and mapping plant locations into the model's regional aggregation. This step is needed to preserve the plant-level heterogeneity that is central to the analysis while ensuring that the sample remains representative at the regional and technological level. We assess the resulting coverage against GSPT and worldsteel benchmarks and use these external controls to verify that the final sample captures the observed global distribution of capacity and production. Appendix B.1 reports the full sample-selection, cleaning, and region-mapping procedure.

PROCESS MODELING AND CARBON ACCOUNTING.—The engineering accounts reported by CRU do not map one-for-one into the model's production structure. We therefore aggregate detailed subprocesses into the model's ironmaking and steelmaking technologies, trace how upstream activities feed into the relevant production routes, and separate captive from merchant iron flows. We also reverse the conventional accounting treatment of process gases and other byproducts as revenue credits, so that carbon accounting follows gross physical energy use rather than net financial cost. This step is essential for measuring embodied emissions consistently across integrated and merchant supply chains. Appendix B.2 provides the full process-modeling and carbon-accounting procedure.

REGIONAL BALANCES.—Next, we reconcile the bottom-up plant data with top-down production and material balances. Plant-level output is reweighted to match worldsteel totals by region and product while preserving within-region heterogeneity. For iron ore, scrap, natural gas, and metallurgical coal, we combine bottom-up input requirements with production and trade statistics to construct physically consistent

[3]The model distinguishes 15 regions: Europe (EUR), US (USA), China (CHN), India (IND), Japan (JPN), South Korea (KOR), Other Asia (OAS), Oceania (OCE), Canada (CAN), Mexico (MEX), Brazil (BRA), Other South America (OSA), Turkey and Other Europe (OEU), CIS and Ukraine (CIS), and Western Asia and Africa (WAF).

regional balances. The purpose is to anchor the benchmark in observed production scales and resource constraints before introducing any counterfactual policy wedges. Appendix B.3 details the balancing procedure.

TRADE FLOW IMPUTATION.—Finally, aggregate customs data must be translated into plant-level trade linkages. We estimate a gravity equation using OECD data to discipline the extensive margin of exporting across region pairs, then assign export participation across plants using a Ricardian ranking by unit costs. By Ricardian ranking, we mean that lower-cost plants are more likely to serve foreign markets and are therefore selected into exporting first. We then reconcile plant-destination flows to observed bilateral trade matrices. This step matters because the policy experiments work through the reallocation of sales across plants, destinations, and positions in the supply chain. Appendix B.4 reports the full trade-imputation procedure.

C. Calibration Strategy and Parameters

CALIBRATION STRATEGY.—We calibrate the model using the calibrated-share-form approach standard in multisector general equilibrium analysis (Rutherford, 1995; Hertel, 1997; van der Mensbrugghe, 2019). The data-construction step delivers the benchmark objects $\{\overline{Q}_{i,a}, \overline{q}_{i,a,f}, \overline{q}_{i,s,a,v}, \overline{\epsilon}_{i,a,v}\}$, which we treat as the reference equilibrium. Plant-level output and engineering accounts pin down benchmark activity levels and nested input shares, while the imputed plant-destination flows pin down the benchmark transformation, sourcing, and trade shares needed to reproduce the observed allocation. Conditional on benchmark prices and policy wedges, the share parameters of the model are chosen so that the numerical equilibrium exactly replicates these benchmark objects.[4]

FREE PARAMETERS.—Conditional on these benchmark shares, the remaining free parameters govern substitution across trade origins, input adjustment within plants, and vertical sourcing between integrated and merchant iron. Table B1 summarizes the full parameterization. Because exact econometric estimates for highly granular intra-plant substitution are scarce, we combine standard macroeconomic values with calibrations grounded in metallurgical constraints. For raw materials, the Armington elasticities (σ_j^A, σ_j^M) follow the GTAP database, with the elasticity among foreign origins twice that between domestic and imported goods, and consumption is Cobb-Douglas (Table B1). For the plant-level nests, elasticities $(\sigma_a, \sigma_a^{\text{KLE}}, \sigma_a^{\text{VA}}, \sigma_a^{\text{Ene}})$ are set low to reflect the stoichiometric and thermal rigidities of heavy industrial production. For instance, substitution in the material nest (σ_a^{Mat}) is tightly bounded in blast furnaces to reflect physical maximums on scrap-to-ore ratios, while operating with greater flexibility in EAFs. Similarly, the vertical integration elasticity (σ_v^{Iron}) is parameterized to reflect the severe logistical and energetic frictions (e.g., thermal penalties) associated with unbundling integrated molten-iron sites.

Beyond the core production nests, the model calibrates several parameters gov-

[4]This differs from the structural inversion exercises common in modern quantitative trade models, where latent productivity or trade-cost objects are recovered from a parsimonious equilibrium system; see, for example, Eaton and Kortum (2002), Anderson and van Wincoop (2003), and Costinot and Rodríguez-Clare (2014). Here a rich benchmark allocation is observed directly, and calibration maps those data into a share system that reproduces the benchmark exactly.

erning supply-side and demand-side heterogeneity. The transformation elasticity (η_a^V) governs the supply-side capability to shift from semi-finished steel into specific product varieties (flat vs. long). On the demand side, the within-domestic variety substitution (σ_v^D) captures how downstream users switch between domestic plants. Finally, factor market rigidities are captured by the price elasticity of factor supply (η_f), distinguishing between the relatively inelastic structural constraints of circular scrap recovery and the highly elastic supply of primary resources, as well as transformation elasticities restricting plant-level mobility of capital and labor (η_f^F). These parameter choices determine how strongly nominal policy wedges are transmitted into changes in production, trade, vertical sourcing, and welfare. To ensure our core findings are not artifacts of these point estimates, we subject the most critical structural and behavioral parameters to an extensive sensitivity analysis in Section V.F.

V. Results

This section uses the calibrated model to quantify, for the global steel industry, how much of the carbon price each design transmits to foreign producers and whom it protects along the supply chain. We first describe the experiments and how they relate to actual policies. We then compare the two designs on global emissions, on who pays and who is protected, and on vertical leakage through intermediates, and turn to resource shuffling, reverse leakage, and sensitivity.

A. Design of Counterfactual Experiments

Each experiment compares a mass-based and a rate-based BCA under the same conditions, so that differences come from the design rather than from stringency. Table 1 summarizes the scenarios.

SETTINGS.— We study two settings. In the EU setting, the border measure accompanies a domestic carbon price; in the US setting, it stands alone. The EU setting is calibrated to the EU but stands for any jurisdiction whose border measure accompanies a domestic carbon price. In both, exporting regions face no carbon price. In addition to the *Global Cap* first-best benchmark and the unilateral *EU Cap*, we compare the two designs in each setting. The *EU M-BCA* extends the EU carbon price τ_{EUR} to the full embodied emissions of imports, while the *EU R-BCA* replaces that charge with a rate-based BCA under the same cap. For each product, the benchmark $\bar{\psi}_{r,v}$ is set to the import intensity in the corresponding mass-based case, so imports are equally emissions intensive under both designs, which is the comparison made in Section II. Trade volumes, credit prices, global emissions, welfare, incidence, and flows of intermediates are determined in equilibrium, and with positive credit prices the rate-based BCA collects no net revenue. In the US, the *US M-BCA* and *US R-BCA* provide the analogous comparison without a domestic carbon price, at a border price equal to the EU carbon price ($87.90 per tCO_2).

CHOICE OF BENCHMARK.—We set the benchmark at the intensity of imports rather than of home production. A benchmark at home intensity would require imports of long steel, pig iron, and DRI to be cleaner than under the mass-based BCA and

Table 1. Counterfactual Scenarios

Scenario	Constraint	Adjusting price	Border price
A. EU Cap	EU steel emissions 30% below baseline	EU permit price	None
B. EU M-BCA	Same cap	EU permit price ($87.90)	$87.90 (EU permit price)
C. EU R-BCA	Same cap; benchmarks equal to EU M-BCA import intensities	EU permit price ($74.94), credit prices	Credit prices
D. EU DM-BCA	Same cap; global emissions equal to EU R-BCA	EU permit price ($76.91), border price	$27.98 ($\lambda = 31.8\%$)
E. US M-BCA	None	None	$87.90 (reference price)
F. US R-BCA	Benchmarks equal to US M-BCA import intensities	Credit prices	Credit prices
G. US DM-BCA	Global emissions equal to US R-BCA	Border price	$10.19 ($\lambda = 11.6\%$)

Notes: Prices in $ per tCO_2. In all scenarios, exporting regions face no carbon price. Benchmarks are set by product; under the rate-based BCA, net receipts are zero. In the discounted scenarios, the border price is the equivalent price, the mass-based price that achieves the global emissions of the rate-based BCA; λ is its ratio to the border price of the mass-based BCA. In the EU DM-BCA, the equivalent price is 0.364 times that scenario's permit price. The Global Cap first-best benchmark is not shown. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

allow imports of flat steel to be dirtier, so the comparison would mix the design with a change in the standard. For pig iron it would be infeasible, because a two-sided tradable standard needs a benchmark within the range of exporters' intensities: EU merchant pig iron, at 1.87 tCO_2 per tonne, is cleaner than every exporter, the cleanest of which emits 2.11 tCO_2 per tonne, so a benchmark at home intensity could be met only by ending pig iron imports.

STRINGENCY.—We select a 30% reduction in EU domestic steel emissions as our primary point of comparison. This stringency level is aligned with the European steel industry's decarbonization pathway under the Fit for 55 package, which aims for a 55% reduction by 2030 relative to 1990 levels—approximately 30% below our 2019 baseline (European Steel Association, 2020). The regime we model is the one after free allocation ends in 2034; Figure 6 reports results for domestic reductions from 5% to 50%.

ACCOUNTING SCOPE AND FACILITY BOUNDARIES.— We measure embodied emissions comprehensively, including direct process emissions, indirect electricity emissions, and emissions from upstream intermediates such as ironmaking. This broader boundary is useful for evaluating BCA design, even though current implementations such as the EU CBAM initially apply narrower coverage. Facility boundaries matter because both rate-based and mass-based designs assess liability using the exporting plant's verified intensity. By modeling trade in both finished steel ($v \in \mathcal{V}_S$) and intermediate iron ($v \in \mathcal{V}_I$), we can quantify vertical leakage under each design (Section II). In the model, resource shuffling is the physical reallocation of output among plants, which verification of plant-level intensities does not prevent.

DISCOUNTED PRICE APPROACH.— Because equal domestic stringency does not im-

ply equal global emissions, we also compare the designs at equal global emissions (Proposition 1). For each rate-based BCA, we solve for the border price at which a mass-based BCA achieves the same global emissions, the *equivalent price*; the discounted mass-based BCA (DM-BCA) applies it. The *equivalent price ratio* λ is the equivalent price divided by the border price of the mass-based BCA. In the EU setting, the equivalent price is \$27.98 per tCO_2 against a border price of \$87.90 under the EU M-BCA, so λ is 31.8%, about a third. In the US setting, the equivalent price is \$10.19 and λ is 11.6%, about an eighth. Welfare is the Hicksian equivalent variation of each region's households and excludes climate damages. Damages cancel when global emissions are equal; when they differ, the net benefit adds the value of the avoided emissions, for which we do not choose a value.

CORRESPONDENCE TO ACTUAL POLICIES.—Once free allocation in the EU ETS has ended in 2034, CBAM corresponds to the mass-based BCA; until then, CBAM charges imports only for emissions not matched by free allocation, which is set by product benchmarks. Border charges are benchmarked because domestic carbon prices are: in the EU until 2034, and permanently where the domestic price works through intensity benchmarks, as in Canada's output-based pricing system and China's national emissions trading system for steel.[5] US proposals benchmark imports against US intensity. Domestic benchmarks are two-sided: a plant cleaner than the benchmark gains, by selling surplus allowances in the EU and Chinese systems or by earning surplus credits in Canada's. A border charge that mirrors them is therefore two-sided as well, which is our rate-based BCA; the one-sided fees in actual proposals, including CBAM's adjustment, which cannot turn negative, deny cleaner importers the gain that cleaner domestic producers receive. The two-sided form also isolates the average charge, and one-sided fees at the same carbon price lie between our two designs on it.

B. Emissions

In both settings, the mass-based design delivers more abatement, whereas the rate-based design softens some price effects but does so by removing the average charge on imports. The equivalent price measures what the average charge is worth in emissions. A mass-based BCA reaches the global emissions of the rate-based BCA at \$27.98 per tCO_2 in the EU setting and \$10.19 in the US setting (rows D and G of Table 2), an equivalent price ratio λ of about a third and about an eighth. The rate-based BCA thus does no more for global emissions than a mass-based BCA at a fraction of its price, although it keeps most of the marginal price.

The rate-based BCA rewards lower intensity nearly as strongly as the mass-based BCA. In the EU setting, its credit price for flat steel is \$81 per tCO_2, against a border price of \$88 under the mass-based BCA, so an exporter that lowers its intensity saves nearly as much under either design. What it removes is the charge on the average tonne imported. Under the mass-based BCA, the average tonne of flat steel imported

[5] For the phase-out of free allocation in the EU, Canada's output-based pricing system, and the allocation of steel allowances in China's national emissions trading system, see ICAP (2026); for the Foreign Pollution Fee Act and the Clean Competition Act, see Elkerbout et al. (2025) and Rennert et al. (2025).

Table 2. Summary of Policy Impacts at Comparable Stringency Levels

Scenario	Global Emissions (%)	(Reverse) Leakage (%)	Domestic Welfare ($bn)	Domestic Output (%) Flat	Long	Macro	Carbon Price ($/tCO$_2$) Domestic	Border
Panel (a): EU Setting								
A. Cap	−0.69	56.44	−0.39	−17.65	−6.68	−0.062	69.71	–
B. M-BCA	−1.34	16.05	−0.89	−16.25	−4.60	−0.126	87.90	87.90
C. R-BCA	−1.02	36.18	−1.38	−17.02	−6.19	−0.081	74.94	81.06 / 71.64
D. DM-BCA	−1.02	36.18	−0.07	−17.12	−5.79	−0.087	76.91	27.98
Panel (b): US Setting								
E. M-BCA	−0.26	(23.05)	−0.88	1.21	1.11	−0.018	–	87.90
F. R-BCA	−0.06	(14.93)	−0.32	0.40	0.86	−0.004	–	67.17 / 83.85
G. DM-BCA	−0.06	(15.72)	−0.01	0.11	0.19	−0.003	–	10.19

Notes: Scenarios are defined in Table 1; points A–G are marked in Figure 6. Changes are relative to the benchmark equilibrium. Leakage in Panel (b), in parentheses, is reverse leakage. Domestic welfare is equivalent variation excluding climate damages. Domestic output is the change in home production of flat and long steel and in the macro sector. The domestic carbon price is the permit price of the cap; the US setting has none. The border price is the mass-based BCA's price and, for the discounted mass-based BCA, the equivalent price; under the rate-based BCA (rows C and F), it is the credit price for flat / long steel; credit prices for pig iron and DRI are reported in Table 3. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

into the EU pays $107 at the border; under the rate-based BCA, charges and credits cancel. Exporters' prices barely differ between the designs, so the difference reaches buyers: the delivered price of flat steel imports is $107 per tonne lower under the rate-based BCA. This is the implicit subsidy embedded in the rate-based design, a subsidy on the average tonne imported (Lemma 1(iii) and Proposition 1(i)).

The subsidy weakens the scale margin as a channel of abatement. Cheaper imports displace home production and add to total demand: for each tonne of EU steel sales that the rate-based BCA displaces, flat steel imports rise by 3.0 tonnes. The additional imports therefore embody more emissions than the home output they replace, which is the condition under which global emissions are higher under the rate-based design (Proposition 1(ii)). Leakage in the EU setting rises from 16.0% under the mass-based BCA to 36.2% under the rate-based BCA. Because the rate-based design leaves the volume of imports unpriced, it is also the costlier way to abate: across the range of global reductions in Figure 5, the mass-based BCA costs less per tonne abated in both settings (Proposition 1(iii)).

The average charge matters more in the US setting. The rate-based BCA achieves 76% of the global emissions reduction of the mass-based BCA in the EU setting, but only 25% in the US setting. The equivalent price ratio mirrors this gap rather than a difference in abatement costs: the less of the mass-based BCA's abatement survives the removal of the average charge, the lower λ. In the US, much of the gap lies in pig iron, which electric arc furnaces import and whose intensities cluster, so that only the average charge reduces its imports (Section V.D): pig iron accounts for 65% of the additional emissions embodied in US imports under the rate-based BCA, against 32% in the EU. Even the mass-based BCA achieves limited global abatement in the US setting, where the cost curves are nearly vertical, partly because US imports are smaller. The two settings also differ in trading partners and technologies, so the

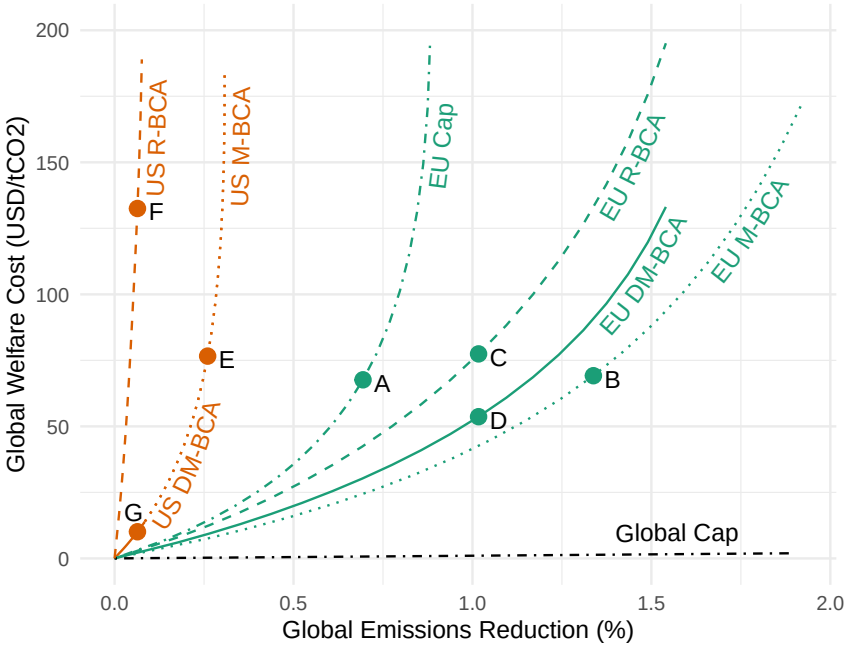


Figure 5. Global Cost-Effectiveness Frontier

Notes: Global welfare cost per tonne of global emissions reduction (USD/tCO_2; welfare excludes climate damages) against global emissions reduction (%). Lines trace policy designs across stringency in the EU and US settings, and labeled points (A–G) mark the comparison points of Table 2. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

comparison between them is not an effect of domestic carbon pricing alone.

C. Who Pays and Who Is Protected

We next turn from environmental performance to economic protection. Figure 6 shows how national welfare changes with stringency under each design. In the EU setting, welfare initially rises at low stringency, by up to \$0.62 billion under the mass-based design and \$0.29 billion under the rate-based design, but this gain disappears as stringency increases. At the comparison point, the rate-based design costs the EU \$1.38 billion, against \$0.89 billion under the mass-based alternative, because it raises no border revenue on net. The mass-based BCA collects \$3.57 billion at the border (Proposition 2(i)); without this revenue, it would be the costlier design for the EU. In the US setting, where the border measure stands alone, the ranking reverses. The rate-based design costs the US \$0.32 billion, against \$0.88 billion under the mass-based alternative, but it also delivers only a quarter as much global abatement (Section V.B).

The rate-based BCA raises nothing on net, but it makes transfers among exporters. Because the credit price clears the benchmark, what plants above the benchmark pay is credited in full to plants below it (Lemma 1(ii)). In the EU setting, \$1.41 billion is transferred from dirtier to cleaner exporters in this way, 40% of it between plants

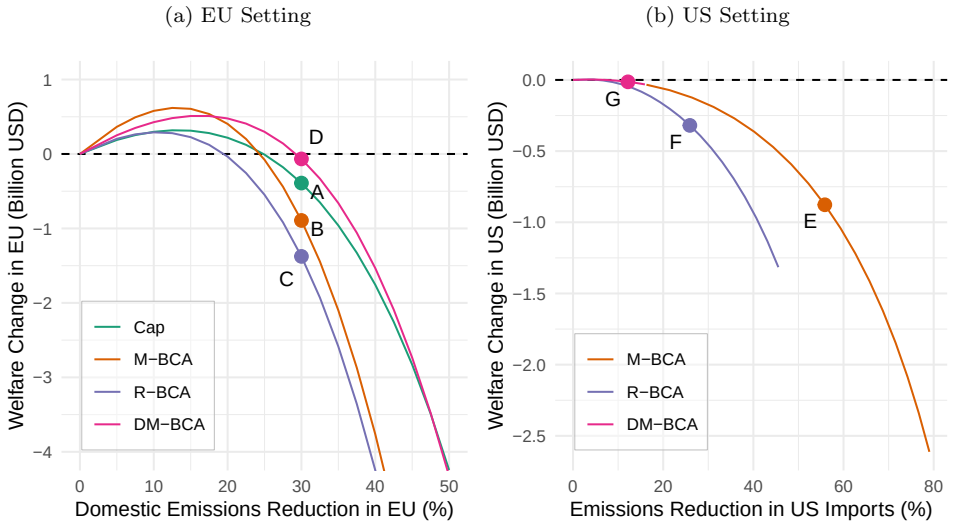


Figure 6. National Welfare Changes

Notes: Panel (a) plots EU welfare change (billion USD) against domestic emissions reduction in the EU (%). Panel (b) plots US welfare change (billion USD) against emissions reduction in US imports (%). Lines trace policy designs across stringency, and labeled points (A–G) mark the comparison points of Table 2. In panel (a), curves continue below the frame at higher stringency. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

in the same exporting region (Table 3). Where the mass-based BCA collects revenue for the importing region, the rate-based BCA makes transfers among and within its trading partners.

At equal global emissions, the mass-based design costs the importing region and the world less. Point D, the discounted mass-based BCA, achieves the same global emissions reduction as the rate-based design at a much lower cost to the EU ($0.07 billion). Of the EU's gain of $1.31 billion, 72% is a gain for the world; the rest comes at the expense of trading partners. In the US setting, the discounted mass-based BCA reduces the US welfare cost to almost zero, and its gain is a gain for the world as a whole. Most of what the importing region gains from the average charge thus reflects a lower cost of abatement, consistent with Proposition 1(iii), and only part of it is a transfer from its trading partners.

Domestically, the mass-based BCA protects home steelmakers more and raises users' prices more. Table 4 shows that the increase in user prices for flat steel is smaller under the rate-based design than under the mass-based alternative (24.2% against 34.6%), because the rate-based design removes the average charge from import prices. The lower input cost is visible in macro outcomes as well: output of the EU macro sector declines by less under the rate-based design (Table 2). At the same time, domestic steel output declines more under the rate-based design; for long steel, the decline is −6.19% against −4.60%, because lower-cost imports exert greater competitive pressure on home producers. This is the industrial incidence in Proposition 2(ii)–(iii): the rate-based design protects downstream users by weakening protection

Table 3. Transfers under the Rate-Based BCA

Product	Benchmark (tCO_2/t)	Credit Price ($/$tCO_2$)	Imports (Mt)	Transfers ($bn) Total	Within Regions
Panel (a): EU Setting, R-BCA					
Pig iron	2.52	55.53	3.50	0.02	0.01
DRI	0.71	363.94	2.15	0.02	0.00
Flat steel	1.22	81.06	24.72	0.97	0.36
Long steel	1.04	71.64	13.65	0.40	0.19
Total				1.41	0.57
Panel (b): US Setting, R-BCA					
Pig iron	2.49	67.08	5.19	0.04	0.03
DRI	0.77	209.93	2.24	0.01	0.00
Flat steel	1.44	67.17	9.27	0.25	0.08
Long steel	0.32	83.85	10.58	0.08	0.02
Total				0.38	0.13

Notes: Rows C and F of Table 2. The benchmark equals the intensity of imports of each product under the mass-based BCA. Transfers are the sum, over exporting plants above the benchmark, of the credit price times the excess intensity times their exports to the importing region; because the credit price clears the benchmark, the same amount is credited to plants below it, and net receipts are zero. Transfers within regions are the part that does not show up in the net positions of exporting regions. R-BCA: rate-based BCA; DRI: direct reduced iron.

Table 4. Import and User Price Changes in the EU

Scenario	Import Price (%) Flat	Long	User Price (%) Flat	Long	Consumer Prices (%)
A. Cap	0.69	−0.26	19.88	8.70	0.02
B. M-BCA	37.10	29.90	34.58	16.36	0.04
C. R-BCA	10.95	5.70	24.18	10.49	0.02
D. DM-BCA	13.80	11.47	25.61	11.85	0.03

Notes: Scenarios are defined in Table 1. Changes are relative to the benchmark equilibrium. Import prices refer to the cost of imported steel products into the EU. User prices refer to the prices paid by steel users within the EU. Consumer prices refer to the overall consumer price index within the EU. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

for home upstream producers.

In the US setting, both designs raise home steel output, more under the mass-based design (1.11% against 0.86% for long steel; Table 2), and the comparison of user prices depends on stringency (Table 5). Against the discounted mass-based BCA, which achieves the same global emissions, the rate-based design cushions user prices at low stringency but raises them more at high stringency. Instrument choice thus governs not just environmental performance, but also how protection is allocated as stringency increases.

The cost abroad falls on a few trading partners (Figure 7). In the EU setting, Western Asia and Africa, China, the CIS region, and Turkey and Other Europe bear the largest losses under both designs. Relative to the mass-based alternative, the rate-based design reduces the losses of the CIS region and the US, while Japan and Korea gain under both designs and Other Asia gains more under the rate-based design. The border charges explain the gains of the CIS region and Other Asia:

Table 5. US Price Impacts of the Rate-Based and Discounted Mass-Based BCAs at Varying Stringency

	Flat Steel (%)		Long Steel (%)		Consumer Prices (%)	
Stringency	R-BCA	DM-BCA	R-BCA	DM-BCA	R-BCA	DM-BCA
Low	0.02	0.38	0.30	0.34	0.000	−0.002
Medium	0.65	1.05	1.60	0.94	0.002	−0.002
High	3.56	1.40	2.95	1.23	0.008	−0.002

Notes: Changes are relative to the benchmark equilibrium. Flat and long steel are user prices in the US. The comparison is between the US rate-based BCA and the discounted mass-based BCA that achieves the same global emissions. The Low, Medium, and High stringency levels correspond to the carbon prices in the EU M-BCA scenario at 10%, 30%, and 50% domestic emissions reduction, respectively ($20, $88, and $241 per tCO_2). M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

Table 6. Pig Iron in the EU

	EU Iron	Captive	Purchased Iron		EU Merchant	Imports	Import Price
Scenario	Production	Use	BOF	EAF	Sales		($/t)
B. M-BCA	62.56	59.12	4.09	1.09	3.45	1.73	544
C. R-BCA	62.07	58.56	5.13	1.61	3.51	3.50	317
D. DM-BCA	62.13	58.62	4.73	1.37	3.51	2.64	392

Notes: Rows B–D of Table 2. Quantities in million tonnes (Mt). EU iron production is pig iron produced in the EU; captive use is iron used in the integrated plant that produces it; purchased iron is merchant pig iron bought by EU steelmakers, from EU merchant sales or imports, by route (BOF: basic oxygen furnace; EAF: electric arc furnace). The import price is the quantity-weighted delivered price. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

under the rate-based design, the CIS region pays $0.25 billion net at the EU border instead of $0.91 billion under the mass-based design, and Other Asia, whose exports are among the cleanest, receives net credits of $0.60 billion. In the US setting, China and the CIS region again lose most, while Canada gains most from the mass-based BCA ($0.50 billion), followed by the EU; under the rate-based design, all of these gains and losses shrink, as does the loss of the US itself.

D. Intermediates and Vertical Leakage

Upstream, the rate-based BCA works differently, because pig iron is a homogeneous intermediate. Its emissions intensity varies much less across exporters than that of long steel (Table C1), reflecting chemical constraints of the blast furnace that limit abatement even at high carbon prices. In our rate-based scenario, benchmarks are set to the emissions intensities of imports observed under the mass-based design, so the pig iron benchmark lies close to the intensity of most exporters. Every exporter's charge or credit is therefore small, not only the average: across exporters, net charges vary by about $16 per tonne, whereas under the mass-based BCA the average imported tonne pays $221, against an import price of about $317 per tonne (Proposition 3(i)). Imported pig iron thus escapes nearly all of the carbon price that EU ironmakers pay.

What follows is vertical leakage through merchant iron. The import price of pig iron is 42% lower under the rate-based design than under the mass-based alternative, and import volumes double (Table 6). Both integrated and electric-arc steelmakers

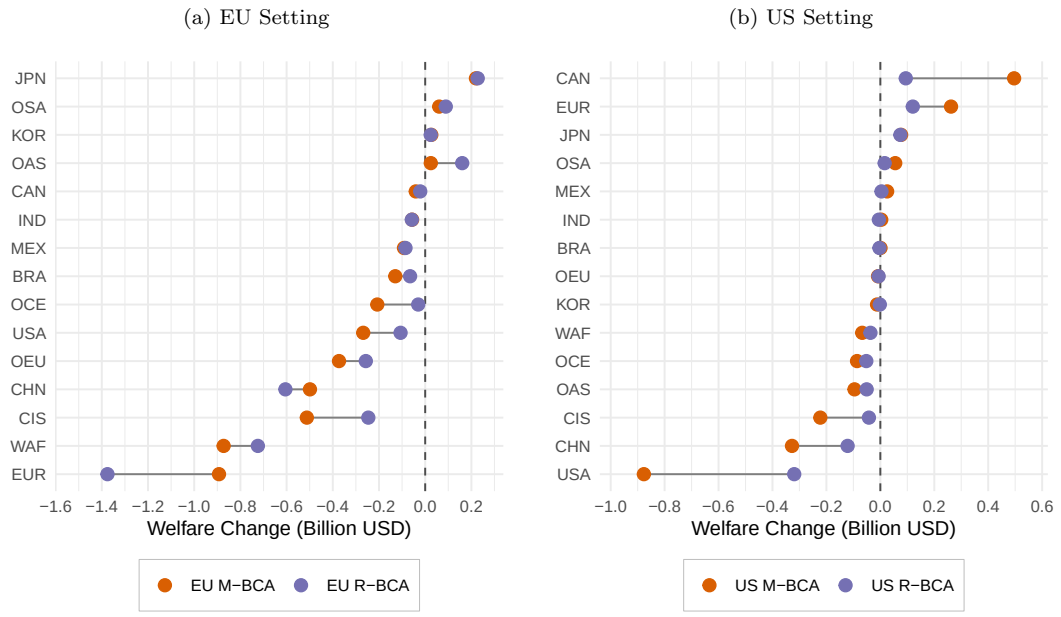


Figure 7. Welfare by Region

Notes: Welfare change (billion USD, equivalent variation excluding climate damages) by region under the mass-based and rate-based BCAs at the comparison point (rows B, C, E, and F of Table 2). Regions are ordered by their welfare change under the mass-based BCA. Region codes are defined in Section IV.A. M-BCA and R-BCA: mass-based and rate-based BCA.

buy more merchant iron, almost all of it imported. Most of the additional imports come from the CIS region, whose pig iron exports to the EU rise from 1.03 Mt to 2.31 Mt (Figure C1). This is the substitution in Proposition 3(ii): with charges close to zero, imported pig iron escapes the domestic carbon price that EU pig iron bears, and downstream steelmakers substitute toward it.

EU ironmaking barely contracts, but foreign merchant ironmaking expands. EU iron production is only 0.8% lower under the rate-based design than under the mass-based alternative. Abroad, the rate-based design more than doubles the increase in emissions relative to the mass-based alternative (22.50 against 9.98 Mt CO_2; Table 7). Emissions from merchant ironmaking rise sharply under the rate-based design, whereas they remain essentially flat under the mass-based design, and merchant ironmaking accounts for 43% of the difference in foreign emissions between the designs.

The result does not depend on where the pig iron benchmark is set: between 0.90 and 1.05 times the matched benchmark, the credit price ranges from \$11 to \$380 per tCO_2, yet pig iron imports stay between 3.26 and 3.51 Mt (Table C2). Its size depends on how easily steelmakers substitute imported for home iron. Integrated BOF plants are optimized for molten-iron charging, cold merchant pig iron entails a substantial thermal penalty, and the global merchant pig iron market remains small relative to total iron production; the model captures these frictions only through the elasticity of substitution between home and imported iron. Halving that elasticity

Table 7. Change in Foreign Emissions by Stage

Scenario	Steelmaking (i)	Ironmaking (ii) On-site	Ironmaking (iii) Merchant	Total
A. Cap	1.46	30.73	2.91	35.11
B. M-BCA	1.29	8.74	−0.04	9.98
C. R-BCA	2.57	14.57	5.36	22.50
D. DM-BCA	1.24	20.04	1.22	22.50

Notes: Rows A–D of Table 2. Change in CO_2 emissions outside the EU relative to the benchmark equilibrium, in million tonnes (Mt): direct steelmaking (i), ironmaking in integrated plants (ii), and merchant ironmaking (iii). M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

Table 8. Decomposition of Foreign Steel and On-Site Ironmaking Emissions

Scenario	Scale	Technique	Composition	Total
A. Cap	26.98	2.17	3.05	32.20
B. M-BCA	8.80	3.29	−2.06	10.03
C. R-BCA	17.24	2.88	−2.97	17.14
D. DM-BCA	17.29	2.66	1.34	21.28

Notes: Rows A–D of Table 2. Change in CO_2 emissions outside the EU from steelmaking and ironmaking in integrated plants (columns (i) and (ii) of Table 7), in million tonnes (Mt), decomposed into scale, composition, and technique effects (Appendix C.3). M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

reduces the additional imports from 1.77 to 1.15 Mt (Table C3).

E. Resource Shuffling and Reverse Leakage

Foreign exporters meet both designs partly through resource shuffling, sending cleaner output to the regulated market (Bushnell, Peterman and Wolfram, 2008). We measure it with a reshuffling index, the ratio of the emissions intensity of exports to the EU to the exporter's domestic average intensity (Table C4). The index for long steel falls from 1.08 in the benchmark equilibrium to 0.55 under both border adjustment designs, and the index for flat steel from 0.85 to 0.53. The reallocation is the same under both designs because it responds to the marginal price, not to the average charge, and the designs are compared at the same intensity of imports (Lemma A1).

Shuffling requires dispersed intensities. The index for pig iron falls only from 1.02 to 0.96, because pig iron intensities cluster across plants (Section V.D). Firms thus exploit technological heterogeneity to satisfy the border measure without necessarily improving global production methods. The reallocation is most pronounced for mixed-technology producers like Japan, where the index for long steel reaches 0.20, effectively diverting the clean capacity of the domestic market to the regulated market. Transfers under the rate-based BCA follow the same pattern: they stay within exporting regions where intensities vary within regions, as for steel, but hardly at all for DRI, whose intensities differ mainly between countries (Table 3).

The designs differ in how much output exporters must reallocate, not in how they reallocate it. Because the rate-based design sustains larger imports, meeting the same

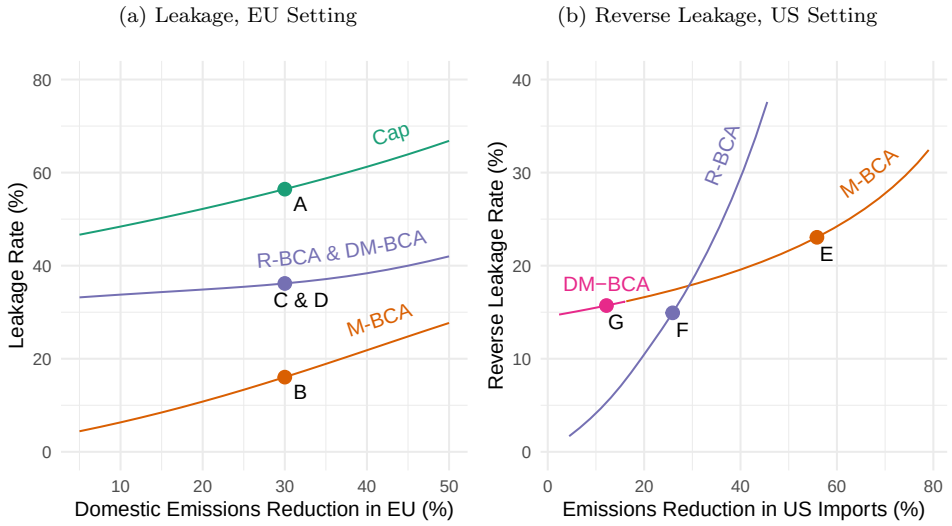


Figure 8. Leakage Dynamics

Notes: Panel (a) plots leakage (%) against domestic emissions reduction in the EU (%). Panel (b) plots reverse leakage (%) against emissions reduction in US imports (%). Lines trace policy designs across stringency, and labeled points (A–G) mark the comparison points of Table 2. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

intensity of imports requires more clean output, and the foreign clean share must rise further (Proposition A1). We decompose the change in foreign emissions from steelmaking and on-site ironmaking into the scale, composition, and technique effects of Antweiler, Copeland and Taylor (2001): changes in total output, in the output shares of technologies, and in emissions intensity within technologies, which sum to the total change (Appendix C.3). The composition effect, the shift toward cleaner technologies, is larger under the rate-based design (Table 8). But the gain is more than offset by the scale effect: larger output abroad adds 17.24 Mt of emissions under the rate-based design, almost double the 8.80 Mt under the mass-based design. The technique effect, which reflects changes in emissions intensity within technologies, is similar across designs and does not drive the ranking.

The US setting provides the mirror image. Figure 8 contrasts leakage in the EU (Panel a) with reverse leakage in the US (Panel b). Consistent with Brunel and Levinson (2026) and Casey, Meng and Rudik (2026), a border measure without a domestic carbon price shifts production home: reverse leakage is 23.0% under the mass-based BCA and 14.9% under the rate-based BCA. By protecting domestic producers without a corresponding domestic carbon cost, the policy increases domestic production and offsets part of the decline in imports. This is the substitution effect in Proposition A2: without a domestic carbon price, the border measure works like a charge on imports alone, which reallocates production toward unabated domestic varieties rather than delivering much abatement abroad.

F. Sensitivity and Limitations

We close by asking whether the central findings are artifacts of the baseline calibration. To answer that question, we vary seven key structural and elasticity parameters between inelastic (Low) and elastic (High) bounds (Appendix C.4). In the EU setting, the mass-based design reduces global emissions by 1.29–1.39% across all variants, compared with 0.96–1.08% for the rate-based design, and the ranges in the US setting are much smaller for both designs. Absolute welfare effects move with substitution elasticities, especially those of the capital, labor, and energy nest and of the top-level production nest, but the ranking does not change: in every variant, the mass-based design reduces global emissions more, the EU is better off under it, and the US is better off under the rate-based design (Table C5 and Figure C2). The mechanisms are similarly stable: EU leakage is higher and EU pig iron imports are roughly twice as large under the rate-based design in every variant (Table C6). Among the seven parameters, pig iron imports respond most to substitution among the material inputs of electric arc furnaces (Figure C3), and less than to the elasticity between home and imported iron (Section V.D). The main message is therefore not a calibration artifact: once the border measure removes the average charge, adjustment shifts toward costlier margins, and a unilateral carbon price is translated less effectively into global abatement.

Four features of the model qualify these results. The plant fleet is fixed: plants adjust output, inputs, and destinations, but there is no entry, exit, or investment in new capacity, so the results describe reallocation among existing plants. Markets are competitive, so the model abstracts from market power in steel and iron. Plant-level trade flows are imputed from regional trade statistics (Appendix B.4), so results for individual exporting plants and regions are less certain than aggregate results. We model the regime after free allocation ends, so neither design is combined with free allocation or export rebates for domestic producers.

VI. Conclusions

This paper studies how the design of a border carbon adjustment determines how much of the carbon price it transmits to foreign producers and whom it protects. In a plant-level multi-region equilibrium model of the global steel industry, we compare mass-based border charges with rate-based benchmarks. As with a domestic intensity standard, a rate-based BCA works as a mass-based BCA combined with a subsidy on the average unit imported: it keeps most of the marginal price but removes the average charge. Without the average charge, the border measure does less for global emissions, and EU leakage rises from 16.0% to 36.2%. Matching the global emissions of the rate-based BCA takes a mass-based BCA at about a third of the border price in the EU setting and an eighth in the US setting.

The average charge also determines the incidence of the border measure. The mass-based BCA raises border revenue for the importing region and protects its steelmakers more; the rate-based BCA raises prices for steel users less, collects nothing on net, and makes transfers from dirtier to cleaner exporters. When the designs achieve the same global emissions, the scaled-down mass-based BCA costs both the importing

region and the world less, although in the EU setting part of the importing region's gain comes at its trading partners' expense.

Because steel combines widely dispersed embodied emissions with trade in carbon-intensive intermediates, it makes visible margins that matter for other emissions-intensive, trade-exposed sectors. Where intensities cluster, as for pig iron, benchmarking leaves imports almost uncharged while domestic producers pay the carbon price, and emissions leak vertically through imported intermediates. The rate-based design then more than doubles EU pig iron imports relative to the mass-based alternative, and foreign merchant ironmaking accounts for 43% of the difference in foreign emissions between the two designs. Resource shuffling, by contrast, needs dispersed intensities and works the same way under both designs.

For a government that wants a milder border measure, the results favor a lower price over a benchmark, setting aside administrative and legal considerations. Where benchmarking is chosen, product-specific design could apply mass-based treatment to homogeneous upstream intermediates while limiting benchmarked treatment to heterogeneous downstream products. One-sided fees, such as CBAM's adjustment, which cannot turn negative, lie between our two designs on the average charge. In the US setting, where the border measure stands alone, both designs do little for global emissions and shift production home, so their main effect is protection rather than abatement abroad.

Four questions remain open. First, with entry, exit, and investment, the designs may also differ in which plants operate, as work on industry dynamics under market-based environmental policy suggests (Fowlie, Reguant and Ryan, 2016). Second, because the designs differ in who pays, they may differ in the strategic responses they invite, such as reciprocal carbon pricing, retaliation, or coalition formation (Nordhaus, 2015; Clausing and Wolfram, 2023; Beaufils, Wanner and Wenz, 2026). Third, with market power, charges and credits may pass through differently along the supply chain, although the rate-based BCA would still remove the average charge. Fourth, how far these mechanisms extend beyond steel depends on the dispersion of emissions intensities, on trade in intermediates, and on whether the border measure accompanies a domestic carbon price, dimensions along which aluminum, cement, chemicals, and other sectors differ (Sogalla, 2026; Stillger, 2025; Clausing et al., 2026).

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Carbon Price Transmission and Economic Protection: The Design of Border Carbon Adjustments

Online Appendix

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A. Proofs and Supporting Results

This appendix provides the proofs of the lemma and propositions in Section II and states two supporting results, on reshuffling and on a border measure without a domestic carbon price.

A.1. Proof of Lemma 1 (Marginal Price and Average Charge)

PROOF: Part (i) follows from the definitions: $dt^M/d\epsilon = \tau^B$ and $dt^R/d\epsilon = \nu$. For part (ii), the charge per unit imported is $t(\epsilon_{f,h})$. Under the mass-based design it equals $\tau^B \epsilon_{f,h} > 0$. Under the rate-based design it equals $\nu(\epsilon_{f,h} - \bar{\psi})$, and complementary slackness requires either $\nu = 0$ or $\epsilon_{f,h} = \bar{\psi}$, so it is zero. Because t is linear in ϵ, the charge per unit imported is the share-weighted average of the charges $t(\epsilon_j)$ on the output of clean and dirty plants, $j \in \{c, d\}$: under the rate-based design, the charges on dirty plants' output equal the credits on clean plants' output. For part (iii), a binding benchmark gives $\epsilon^R_{f,h} = \bar{\psi} = \epsilon^M_{f,h}$. By part (ii), the charge per unit imported is then $\tau^B \bar{\psi}$ under the mass-based design and zero under the rate-based design, for any $\nu > 0$.

A.2. Proof of Proposition 1 (Emissions)

PROOF: For part (i), imports have the same intensity $\bar{\psi}$ under both designs, and hence the same clean share $s_{c,f,h}$. By Lemma 1, $p^M_{f,h} = c(s^M_{c,f}) + \tau^B \bar{\psi}$ and $p^R_{f,h} = c(s^R_{c,f})$, where the regional clean share $s_{c,f} = \varphi_{f,h} s_{c,f,h}$ rises with the volume of imports. Suppose $p^R_{f,h} \geq p^M_{f,h}$. The home price $p_{h,h}$ does not depend on the border design, and demand for the foreign variety decreases in its own price, so $q^R_{f,h} \leq q^M_{f,h}$, hence $s^R_{c,f} \leq s^M_{c,f}$ and $p^R_{f,h} = c(s^R_{c,f}) \leq c(s^M_{c,f}) < p^M_{f,h}$, a contradiction. Therefore $p^R_{f,h} < p^M_{f,h}$ and $q^R_{f,h} > q^M_{f,h}$, and because P increases in $p_{f,h}$, $P^R < P^M$ and $Q^R > Q^M$.

For part (ii), global emissions are $E = \epsilon_{h,h} q_{h,h} + \epsilon_{f,h} q_{f,h} + \epsilon_{f,f} q_{f,f}$. The intensities and $q_{f,f}$ are the same under both designs, so $E^R - E^M = \bar{\psi}(q^R_{f,h} - q^M_{f,h}) + \epsilon_{h,h}(q^R_{h,h} - q^M_{h,h})$, which is positive if the emissions embodied in the additional imports exceed those of the home output displaced.

For part (iii), let E^R denote global emissions under the rate-based design. With $\tau_h = \tau^B$ and inelastic foreign domestic demand, a mass-based design puts the same price on every tonne whose emissions respond to policy: those from the clean shares in both regions, from the mix of home and imported varieties, and from the scale of Q. Each of these margins is therefore adjusted until its marginal cost per tonne equals the price, which is the condition for achieving a given level of emissions at least cost; by continuity, some mass-based price achieves E^R in this way. The rate-based design puts no charge on the average unit imported, so the marginal value of imports falls short of their marginal emissions cost. Unless the volume of imports does not respond to this wedge, the rate-based allocation differs from the least-cost allocation that achieves E^R and therefore costs more.

A.3. *Proof of Proposition 2 (Who Pays and Who Is Protected)*

PROOF: Part (i) follows from Lemma 1(ii): border revenue is the charge per unit imported times $q_{f,h}$, and under the rate-based design the charges on dirty plants' output and the credits on clean plants' output cancel.

For part (ii), the home price $p_{h,h} = c(s_{c,h}) + \tau_h \epsilon_{h,h}$ does not depend on the border design, while $p^M_{f,h} > p^R_{f,h}$ by Proposition 1(i). Because F is linearly homogeneous, the ratio $q_{h,h}/q_{f,h}$ depends only on the price ratio $p_{h,h}/p_{f,h}$ and decreases in it, so home producers' share is larger under the mass-based design.

For part (iii), the surplus of downstream firms is $\Pi_d = \max_Q\{G(Q) - PQ\}$, the value-added accruing to fixed factors in downstream production. By the Envelope Theorem:

$$\frac{d\Pi_d}{dP} = -Q(P) < 0 \tag{A1}$$

Since the dual price index P is increasing in input prices, $P(p_{h,h}, p^M_{f,h}) > P(p_{h,h}, p^R_{f,h})$. Therefore, downstream surplus is strictly higher under the rate-based design:

$$\Pi_d(P^R) > \Pi_d(P^M) \tag{A2}$$

A.4. *Proof of Proposition 3 (Intermediates and Vertical Leakage)*

PROOF: For part (i), the rate-based charge on a unit from a plant with intensity ϵ_j is $\nu(\epsilon_j - \bar{\psi})$, whose absolute value is at most $\nu\delta$; home producers pay τ_h on their emissions, and under the mass-based design the charge per unit imported is $\tau^B\bar{\psi}$ by Lemma 1(iii).

For part (ii), the price of the intermediate for a domestic producer is $p_{h,h} = c(s_{c,h}) + \tau_h\epsilon_{h,h}$. For a foreign producer, the price inclusive of the border charge is $p_{f,h} = c(s_{c,f}) + t(\epsilon_{f,h})$. Under the rate-based design, $|t(\epsilon_{f,h})| \leq \nu\delta$ by part (i). If production costs are approximately equal across regions, the resulting price wedge is

$$p_{h,h} - p_{f,h} \approx [c(s) + \tau_h\epsilon_{h,h}] - [c(s) + 0] = \tau_h\epsilon_{h,h}, \tag{A3}$$

so the domestic producer faces a competitive disadvantage close to the full domestic carbon cost. Because F is linearly homogeneous, the ratio $q_{f,h}/q_{h,h}$ increases in $p_{h,h}/p_{f,h}$, so downstream firms substitute toward imported intermediates.

A.5. *Supporting Results*

Foreign producers can lower the intensity of their exports by raising their clean share or by sending the output of clean plants to the regulated market and that of dirty plants elsewhere, a response known as resource shuffling (Bushnell, Peterman and Wolfram, 2008).

LEMMA A1 (Reshuffling Priority): *Under either design with a positive marginal price, foreign producers allocate the output of clean plants to the regulated market*

first, because each clean unit sent there instead of a dirty unit lowers the border charge by the marginal price times the intensity gap $\epsilon_d - \epsilon_c$.

PROOF: The foreign producer allocates the output of its clean and dirty plants, $\{q_c, q_d\}$, to two markets $\{h, f\}$. The producer minimizes total charge-inclusive costs:

$$\min \sum_{j \in \{h,f\}} [c(s_{c,j})q_j + t(\epsilon_j)q_j \cdot \mathbb{I}_{j=h}] \tag{A4}$$

subject to $\sum_j q_{c,j} \leq q_c$ and $\sum_j q_{d,j} \leq q_d$. The first-order condition for allocating a clean unit to market h instead of a dirty unit yields a charge saving equal to the marginal price times $(\epsilon_d - \epsilon_c)$, which is strictly positive under both designs (τ^B or ν). Since costs are otherwise identical across markets, the producer allocates clean output to the regulated market h first.

PROPOSITION A1 (Reshuffling and Clean Output): *For a given intensity of imports* $\bar{\psi}_{f,h}$*, the quantity of clean plants' output that must be sent to the regulated market rises in proportion to the volume of imports. Because the rate-based design sustains higher imports* $q_{f,h}$ *(Proposition 1), it requires more clean output; once reallocating existing clean output between markets is exhausted, the foreign region's clean share* $s_{c,f}$ *must rise.*

PROOF: Consider a comparison between a rate-based and a mass-based design at the same intensity of imports. Let $q_f = q_{f,h} + q_{f,f}$ be the total foreign production. By Lemma A1, producers prioritize the allocation of clean units to the regulated market. The observed intensity at the border, $\epsilon_{f,h}$, is the weighted average of the intensities of the clean ($q_{c,f,h}$) and dirty ($q_{d,f,h}$) units allocated to the export market:

$$\epsilon_{f,h} = \frac{q_{c,f,h}\epsilon_c + (q_{f,h} - q_{c,f,h})\epsilon_d}{q_{f,h}} \tag{A5}$$

Solving for the volume of clean units required to hit the target $\bar{\psi}_{f,h}$ yields:

$$q_{c,f,h} = q_{f,h}\left[\frac{\epsilon_d - \bar{\psi}_{f,h}}{\epsilon_d - \epsilon_c}\right] \tag{A6}$$

From Proposition 1(i), the rate-based design results in a lower price for the variety sold in the regulated market ($p^R_{f,h} < p^M_{f,h}$), which sustains a larger export volume $q^R_{f,h} > q^M_{f,h}$. At this higher export scale, the existing clean output $s_{c,f}q_f$ is spread over a larger volume, making it more difficult to achieve the intensity target $\bar{\psi}_{f,h}$ through reshuffling alone.

To maintain the target $\bar{\psi}_{f,h}$ at the larger scale $q^R_{f,h}$, the required absolute clean production $q^R_{c,f,h}$ must satisfy:

$$q^R_{c,f,h} = q^R_{f,h}\left[\frac{\epsilon_d - \bar{\psi}_{f,h}}{\epsilon_d - \epsilon_c}\right] > q^M_{f,h}\left[\frac{\epsilon_d - \bar{\psi}_{f,h}}{\epsilon_d - \epsilon_c}\right] = q^M_{c,f,h} \tag{A7}$$

Since the total volume of clean units must satisfy $s_{c,f}q_f \geq q_{c,f,h}$, the larger requirement $q^R_{c,f,h}$ necessitates a higher regional clean share $s_{c,f}$ in the foreign sector, that is, more output from clean plants. Thus, by preserving trade scale, the rate-based design requires more clean output for the same intensity of imports.

Reshuffling depends on the marginal price, not on the average charge, so it works the same way under both designs, and it requires dispersed intensities: if $\epsilon_c = \epsilon_d$, the intensity of imports cannot change.

PROPOSITION A2 (Reshoring and Diluted Abatement): *In the US setting ($\tau_h = 0$), the border measure induces reshoring of domestic production. The net impact on global emissions is determined by the trade-off between the abatement effect in the foreign sector and the substitution effect toward dirtier domestic varieties. Global emissions reductions are diluted if the reshoring of unpriced domestic production is large relative to the clean-share response of foreign producers.*

PROOF: Total global emissions are defined as $E_{global} = \epsilon_{h,h}q_{h,h} + \epsilon_{f,h}q_{f,h} + \epsilon_{f,f}q_{f,f}$, where $\epsilon_{f,h}$ and $\epsilon_{f,f}$ denote the average intensities of foreign production allocated to the regulated and unregulated markets, respectively. Differentiating with respect to the border charge t:

$$\frac{dE_{global}}{dt} = \underbrace{(\epsilon_{h,h} - \epsilon_{f,h})\frac{dq_{h,h}}{dt}}_{\text{Substitution Effect}} + \underbrace{q_{f,h}\frac{d\epsilon_{f,h}}{dt} + q_{f,f}\frac{d\epsilon_{f,f}}{dt}}_{\text{Abatement/Reshuffling Effect}} \tag{A8}$$

With $\tau_h = 0$, domestic intensity $\epsilon_{h,h}$ is constant. Since the BCA raises the price of the foreign variety, $\frac{dq_{h,h}}{dt} > 0$ (reshoring). If domestic production is dirtier than the foreign imports ($\epsilon_{h,h} > \epsilon_{f,h}$), the substitution effect is positive. While the foreign producer may clean up its exports ($\frac{d\epsilon_{f,h}}{dt} < 0$), if this cleanup is achieved primarily through reshuffling existing units from the domestic to the export market rather than by a higher regional clean share ($s_{c,f}$), then $d\epsilon_{f,f}/dt > 0$, and the net global abatement is diminished.

B. Data Construction

This appendix details the construction of the global dataset, which harmonizes plant-level engineering estimates with aggregate trade and production statistics. The procedure ensures that the micro-level heterogeneity in technology and efficiency is preserved while satisfying macro-level constraints on global supply and demand.

B.1. Data Sources and Sample Selection

The core iron and steel production data is drawn from the CRU Steel Cost Service for the baseline year 2019 (CRU Group, 2019). We extract key physical and economic parameters for each facility, including production capacities, operating rates, granular material inputs (e.g., kg per tonne of output), energy consumption vectors, and local unit costs for factors of production.

We apply a rigorous data cleaning and validation protocol to clean and harmonize the raw data. Plants are excluded if they report discontinuous operations, missing vital cost components, or negligible sales volumes (defined as less than 0.1 kt annually). Following the plant-level filtering, we map individual ISO-alpha3 country codes from plant locations into the model's distinct custom regions (e.g., USA, China, EU28/EFTA, Russia/CIS/Ukraine, and rest of world aggregates). This mapping balances computational tractability with the resolution required for accurate policy evaluation. We validate the representativeness of this cleaned sample by cross-referencing aggregate regional capacities against the Global Steel Plant Tracker (GSPT) and production totals against the World Steel Association (worldsteel) (Global Energy Monitor, 2023; World Steel Association, 2022).

B.2. Process Modeling and Carbon Accounting

Translating engineering cost accounts into economic production functions requires two major structural adjustments. First, we map granular sub-processes into the model's aggregated sectors. We incorporate the physical and economic profiles of upstream activities—specifically blast furnace pelletizing, direct reduction (DR) pelletizing, coke making, and sintering—and consolidate their costs and physical inputs recursively into the primary ironmaking functions (blast furnace, DRI, Corex, and custom ironmaking). Specific configurations, such as Conarc and custom iron or steelmaking setups, are standardized into representative blast furnace or DRI functions. Crucially, the pipeline calculates net intermediate production to maintain mass-balance consistency. Pig iron and DRI outputs are dynamically split between captive consumption (used on-site for steelmaking) and merchant surplus (sold to third parties).

Second, we rigorously correct for byproducts to ensure accurate carbon accounting. Standard industry cost accounting frameworks typically treat process gases (e.g., blast furnace gas, coke oven gas) and surplus generated power or steam as revenue credits that artificially lower net operational costs. For a physical climate model, however, these must be treated as conserved energy flows to track emissions accurately. To resolve this, we calculate a mathematical correction factor ϕ^{bp} to convert

these financial credits back into gross physical input requirements:

$$\phi^{bp} = 1 - \frac{\text{Total Production}}{\text{Total Consumption}} \tag{B1}$$

Here, "Total Consumption" refers to the absolute gross amount of an energy carrier a plant physically requires for its processes, while "Total Production" is the volume of that carrier the plant generates internally as a byproduct. The factor ϕ^{bp} calculates the share of the required energy that must be purchased externally. Multiplying the gross input requirement by ϕ^{bp} yields the net physical external input, preventing the double-counting of emissions from self-generated process gases and ensuring the model captures the true energy intensity of production. We also calculate and add the auxiliary energy requirements for desulfurization and secondary steelmaking to the core process definitions to capture the full, unabated energy footprint of semi-finished steel products.

B.3. Regional and Raw Material Balances

To reconcile the bottom-up microdata with official national accounts, we employ a systematic re-weighting procedure. Plant-level output is scaled to match authoritative worldsteel regional control totals for both intermediate inputs (pig iron, DRI) and final steel goods (flat and long products) (World Steel Association, 2022). We calculate a regional adjustment factor $\alpha_{r,s}$ for each region r and sector s:

$$\alpha_{r,s} = \frac{Y_{r,s}^{\text{worldsteel}}}{Y_{r,s}^{\text{CRU}}} \tag{B2}$$

where Y denotes aggregate production. Each plant's output is multiplied by this factor, ensuring that the baseline simulation exactly reproduces observed regional distinctives in production scale without altering the underlying micro-level cost distributions or technological shares.

For primary raw materials, we construct physically consistent global balances with differing approaches depending on data availability. For iron ore and ferrous scrap, total consumption (C) is determined bottom-up from CRU plant requirements, while bilateral trade flows (exports X and imports M) are anchored to UN Comtrade statistics (United Nations Statistics Division, 2024). Production (Y) is therefore derived as the residual ($Y = C + X - M$), with imports proportionally scaled down if they implausibly exceed total consumption. Missing Comtrade quantities are imputed using regional average prices. Natural gas bilateral trade (pipeline and liquefied natural gas) and production are sourced directly from the internally consistent BP Statistical Review (BP, 2022), with consumption calculated as the residual.

In contrast to gas and ore, the dataset isolates the trade of metallurgical (coking) coal rather than thermal coal. Regional production and consumption data for metallurgical coal are sourced from the U.S. Energy Information Administration (EIA) (U.S. Energy Information Administration, 2024), but consistent bilateral trade flows are unavailable at the same coverage. Therefore, we assume bilateral flows follow the

general coal trade patterns (HS code 270112) from UN Comtrade (United Nations Statistics Division, 2024). To reconcile the discrepancies between the independent macroeconomic totals and the trade network, we frame the problem as a quadratic programming optimization. This minimizes the squared deviation from the reported EIA totals subject to the global material balance constraint ($Y_r + M_r = C_r + X_r$, and $\sum_r X_r = \sum_r M_r$). After finding the optimal balanced macro totals, we apply Iterative Proportional Fitting (IPF) to adjust the Comtrade bilateral matrix to exactly match these new targets.

B.4. Trade Flow Imputation

Modeling trade with heterogeneous firms requires determining not just how much is traded between countries (intensive margin), but which specific plants participate in export markets (extensive margin). Since customs data (UN Comtrade) provides only aggregate bilateral flows (United Nations Statistics Division, 2024), we impute plant-level trade linkages for final steel products (flat and long steel) through a three-step procedure that combines structural gravity estimation with our micro-level cost data. Intermediate iron products (pig iron and DRI) are allocated proportionally based on plant-level surplus production.

CALIBRATION OF EXPORTING MARGINS.—First, we calibrate the extensive margin—the share of firms in a region that export to a specific destination—using firm-level trade statistics from the OECD. We integrate multiple OECD datasets: Inter-Country Input-Output (ICIO) tables for trade flows (X_{ijs}), International Transport and Insurance Costs (ITIC) for transport margins between cost, insurance, and freight (CIF) and free on board (FOB) values, Structural and Demographic Business Statistics (SDBS) for total firm counts (N_{iis}), and Trade by Enterprise Characteristics (TEC) for exporting firm counts (N_{is}) (OECD, 2021, 2024*a*,*b*,*c*). Following Sogalla, Wanner and Watabe (2024), we estimate a gravity regression to predict the destination-specific exporting firm share s_{ijs}:

$$\ln(s_{ijs}) = \beta_0 + \beta_1 \ln(\chi_{ijs}) + \beta_2 \ln(\text{dist}_{ij}) + \gamma_j + u_{ij} \tag{B3}$$

where $\chi_{ijs} = X_{ijs}/X_{iis}$ is the bilateral trade intensity, $\text{dist}_{ij} = 1/(1 - \text{ITIC}_{ij})$ is the transport cost friction, and γ_j are destination fixed effects. This estimation provides a region-pair specific target for the proportion of capacity that serves each export market.

RICARDIAN SELECTION.—Second, we identify the specific plants that fill these export quotas using a Ricardian selection mechanism. Consistent with trade theory, we assume that more productive (lower-cost) firms are more likely to overcome fixed trade costs. Within each region, we rank plants by their unit production costs. We determine the discrete plant count quota for exports as $\lceil s_{ijs} N_{is} \rceil$. We then assign export status to the lowest-cost plants until this quota is satisfied for each destination. This ranking mechanism ensures that the model preserves the theoretical correlation between productivity and export participation observed in microdata.

BILATERAL FLOW RECONCILIATION.—Finally, we determine the specific volume of trade for each plant-destination pair. Initial export volumes are allocated based on a

gravity-style share equation where market share is inversely related to delivered costs (production plus transport), utilizing a constant elasticity parameter. Because these initial allocations do not sum perfectly to observed national trade totals, we apply an IPF method to iteratively adjust the flows. This parallelized algorithm scales the plant-level exports until they simultaneously satisfy two constraints: (1) the sum of all exports and domestic sales from a plant exactly equals its modeled production capacity, and (2) the aggregation of plant-level flows between two regions exactly matches the harmonized bilateral trade volume reported in UN Comtrade.

B.5. Calibration, Plant Data, and Trade

Table B1 lists the calibrated parameters, Figure B1 shows the distribution of plant capacities in the CRU and GSPT data, and Table B2 reports the steel trade of the EU and the US in the benchmark year.

Table B1. Calibrated Parameters

Panel (a): Production Substitutions and Transformations

Symbol	Description	Value
Steel Production		
σ_a	Top-level	0.1
σ_a^{KLE}	KLE nest	0.4 (BF/DRF), 0.5 (BOF/EAF)
σ_a^{VA}	Value-added nest	1.0
σ_a^{Ene}	Energy nest	0.1 (BF/DRF), 0.2 (BOF/EAF)
σ_a^{Mat}	Material nest	0.3 (BOF), 0.4 (BF/DRF), 2.5 (EAF)
η_a^V	Transformation	0.5
σ_v^{Iron}	Iron make-or-buy	0.1 (Pig iron), 0.5 (DRI)
Macro Production		
σ^C	Top-level	1.0
$\sigma_{\text{mac}}^{\text{VA}}$	Value-added nest	1.255
$\sigma_{\text{mac}}^{\text{Mat}}$	Material nest	0.3

Panel (b): Trade and Final Demand Elasticities

Commodity	Domestic vs. Imported (σ^A)	Across Foreign Origins (σ^M)	Within Domestic (σ^D)
Flat steel	1.4	5.0	5.0
Long steel	1.4	8.0	8.0
Pig iron	1.4	10.0	10.0
DRI	1.4	10.0	10.0
Iron ore	0.9	1.8	–
Steel scrap	2.95	5.9	–
Coal	3.05	6.1	–
Natural gas	10.0	20.0	–
Macro good	2.46	4.92	–

Panel (c): Factor Supply and Mobility Elasticities

Factor (f)	Factor Supply (η_f)	Plant Mobility (η_f^F)
Capital	–	0.1
Labor	–	2.0
Natural gas	2.0	2.9
Electricity	4.0	4.2
Iron ore	10.0	4.5
Coal	10.0	5.8
Steel scrap	0.5	13.8

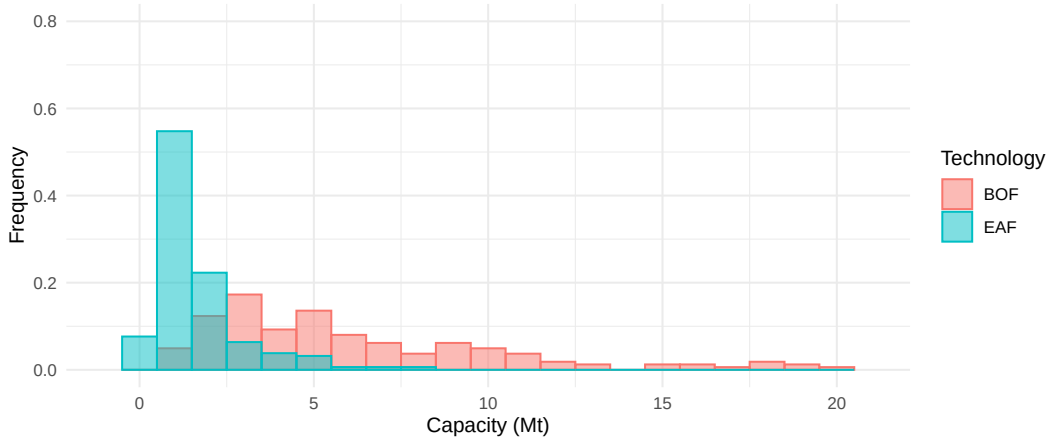
Notes: BF: blast furnace; DRF: direct reduction furnace; BOF: basic oxygen furnace; EAF: electric arc furnace; DRI: direct reduced iron. KLE: capital, labor, and energy; VA: value added.

Table B2. Steel Trade of the EU and the US (2019)

Region	Production (Mt)	Consumption (Mt)	Imports (Mt)	Import Share (%)	Main Origins of Imports (Mt)
EU	159.4	163.7	34.0	20.8	OEU (9.1), CIS (9.0), KOR (3.9)
US	87.8	99.9	20.1	20.2	CAN (4.9), EUR (4.1), KOR (2.2)

Notes: Consumption is apparent consumption: domestic production plus imports minus exports. The import share is imports as a share of consumption. Main origins are the three largest exporting regions; region codes are defined in Section IV.A. The EU comprises the EU28 and EFTA countries. Source: Own calculations based on worldsteel and UN Comtrade data.

(a) CRU Steel Cost Service (Number of Plants: 347)



(b) Global Steel Plant Tracker (Number of Plants: 932)

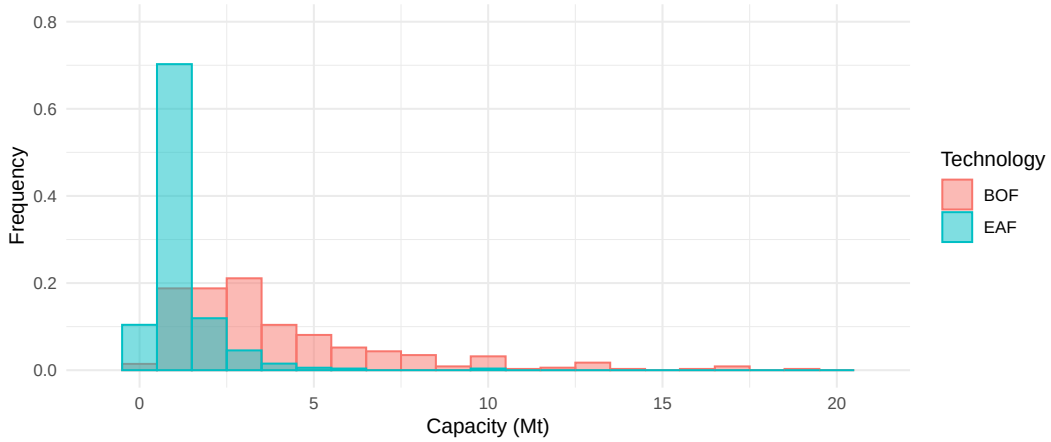


Figure B1. Histogram of Steel Plants by Production Capacity per Year

Notes: Panels show the distribution of plant capacities; sample sizes are reported in the panel captions. BOF: basic oxygen furnace; EAF: electric arc furnace. Source: CRU Steel Cost Service and Global Steel Plant Tracker.

C. Additional Quantitative Results

C.1. Markets and Trade

Table C1 reports quantities, emissions intensities, and delivered prices of pig iron and long steel in the EU market, and Figure C1 EU imports by product and exporting region.

Table C1. Pig Iron and Long Steel in the EU Market

Scenario	EU Producers (EU → EU)			Imports (Foreign → EU)		
	Quantity (Mt)	Intensity (tCO_2/t)	Price ($/t)	Quantity (Mt)	Intensity (tCO_2/t)	Price ($/t)
Panel (a): Pig Iron						
Benchmark	4.93	2.12 (0.33)	351 (43)	3.03	2.67 (0.37)	317 (26)
Cap	3.50	1.90 (0.27)	487 (64)	3.41	2.67 (0.36)	317 (25)
M-BCA	3.45	1.87 (0.26)	520 (69)	1.73	2.52 (0.32)	544 (37)
R-BCA	3.51	1.90 (0.26)	497 (65)	3.50	2.52 (0.31)	317 (31)
DM-BCA	3.51	1.89 (0.26)	500 (66)	2.64	2.60 (0.34)	392 (28)
Panel (b): Long Steel						
Benchmark	60.71	0.58 (0.79)	449 (41)	12.11	2.07 (1.01)	401 (69)
Cap	58.54	0.44 (0.61)	490 (52)	13.32	2.06 (1.01)	400 (69)
M-BCA	60.47	0.42 (0.58)	504 (56)	10.59	1.04 (1.05)	487 (101)
R-BCA	58.88	0.43 (0.60)	494 (52)	13.65	1.04 (1.01)	388 (87)
DM-BCA	59.36	0.43 (0.60)	495 (53)	11.83	1.69 (1.12)	444 (78)

Notes: Rows A–D of Table 2 and the benchmark equilibrium. Quantity-weighted mean and standard deviation (in parentheses) across plants of the emissions intensity and the delivered price, for EU producers' sales in the EU and for imports into the EU. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA.

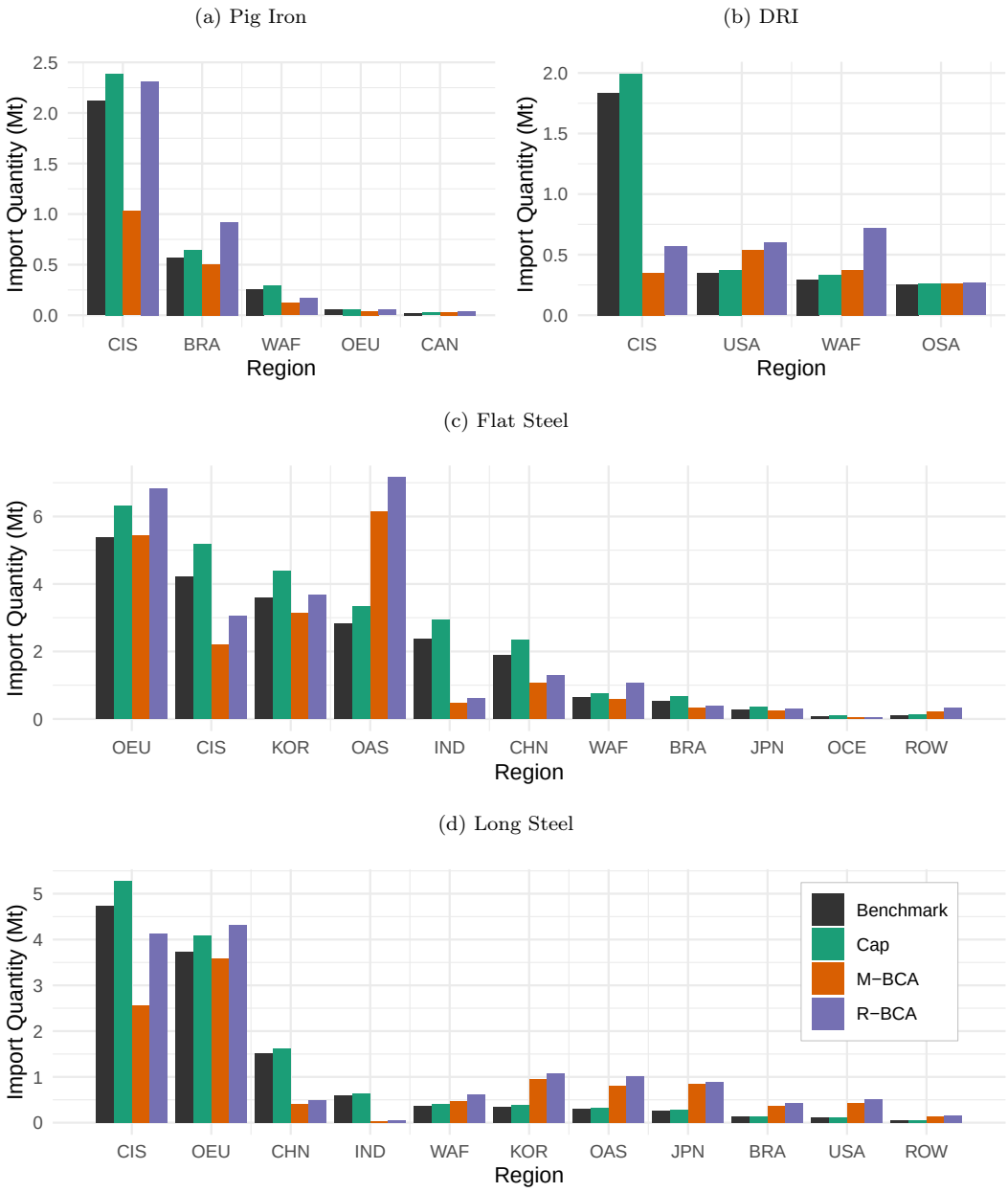


Figure C1. EU Imports by Product and Exporting Region

Notes: Imports into the EU (Mt) by exporting region in the benchmark equilibrium and in rows A–C of Table 2. For flat and long steel, exporters beyond the ten largest in the benchmark are grouped as ROW. Region codes are defined in Section IV.A. M-BCA and R-BCA: mass-based and rate-based BCA; DRI: direct reduced iron.

C.2. Intermediates

Table C2 varies the pig iron benchmark under the EU rate-based BCA, and Table C3 the upstream elasticities of substitution (Section V.D).

Table C2. Pig Iron Benchmark under the EU Rate-Based BCA

Pig Iron Benchmark	Benchmark (tCO_2/t)	Credit Price ($/$tCO_2$)	Pig Iron Imports (Mt)	Global Emissions (%)	Domestic Welfare ($bn)	Leakage (%)
0.88× matched	2.21	851.52	3.03	−1.04	−1.58	35.04
0.90× matched	2.26	379.53	3.26	−1.03	−1.50	35.50
0.95× matched	2.39	131.70	3.44	−1.02	−1.41	35.95
Matched	2.52	55.53	3.50	−1.02	−1.38	36.18
1.05× matched	2.64	10.86	3.51	−1.01	−1.36	36.37

Notes: EU R-BCA (row C of Table 2) with the pig iron benchmark scaled relative to the matched benchmark, the intensity of pig iron imports under the EU M-BCA; benchmarks for the other products are unchanged. Global emissions, domestic welfare, and leakage are as in Table 2. M-BCA and R-BCA: mass-based and rate-based BCA.

Table C3. Upstream Elasticities

Elasticity	EU Pig Iron Imports (Mt)			EU R-BCA		US R-BCA	
	M-BCA	R-BCA	R − M	Leakage (%)	Welfare ($bn)	Emissions (%)	Leakage (%)
Baseline	1.73	3.50	1.77	36.18	−1.38	−0.06	14.93
Iron σ^D, σ^M × 0.5	1.72	3.51	1.79	36.38	−1.37	−0.06	16.64
Iron σ^D, σ^M × 0.25	1.71	3.51	1.80	36.50	−1.37	−0.05	17.27
Natural gas σ^A, σ^M × 0.5	1.73	3.50	1.77	36.22	−1.38	−0.06	14.83
Iron σ^A × 0.5	1.84	2.99	1.15	34.95	−1.36	−0.06	14.24
Iron σ^A × 0.25	1.90	2.75	0.85	34.34	−1.35	−0.06	13.90

Notes: Each row scales trade elasticities relative to the baseline calibration (Table B1): for pig iron and DRI, among domestic plants and among import origins (σ^D, σ^M) or between home and imported iron (σ^A); for natural gas, between home and imported gas and among import origins (σ^A, σ^M). Imports are EU pig iron imports under the EU M-BCA and R-BCA (rows B and C of Table 2), and R − M is their difference; EU leakage and domestic welfare are for the EU R-BCA; global emissions and reverse leakage are for the US R-BCA (row F). M-BCA and R-BCA: mass-based and rate-based BCA; DRI: direct reduced iron.

C.3. Resource Shuffling

Table 8 decomposes the change in emissions from steelmaking and on-site iron-making outside the EU. Following the identity $E = \sum_a Y s_a e_a$, where Y is total production, $s_a = Y_a/Y$ is the production share of technology a, and $e_a = E_a/Y_a$ is its emissions intensity, we isolate three effects:

$$\Delta E = \underbrace{\sum_a w_a \ln\left(\frac{Y^1}{Y^0}\right)}_{\text{Scale Effect}} + \underbrace{\sum_a w_a \ln\left(\frac{s_a^1}{s_a^0}\right)}_{\text{Composition Effect}} + \underbrace{\sum_a w_a \ln\left(\frac{e_a^1}{e_a^0}\right)}_{\text{Technique Effect}}, \tag{C1}$$

where $w_a = (E_a^1 - E_a^0)/(\ln E_a^1 - \ln E_a^0)$ is the logarithmic mean weight, and superscripts 0 and 1 denote the benchmark equilibrium and the policy scenario. With these weights the three effects sum exactly to the total change.

Table C4 reports the reshuffling index by product (Section V.E).

Table C4. Reshuffling Index by Product

Product	Benchmark	A. Cap	B. M-BCA	C. R-BCA	D. DM-BCA
Pig Iron	1.02	1.02	0.96	0.96	1.00
DRI	0.70	0.70	0.66	0.66	0.69
Flat Steel	0.85	0.86	0.53	0.53	0.74
Long Steel	1.08	1.09	0.55	0.55	0.90

Notes: Emissions intensity of exports to the EU relative to the exporters' average intensity, for all exporters together, in the benchmark equilibrium and in rows A–D of Table 2. Values below 1 mean that exporters send cleaner-than-average output to the EU. M-BCA, R-BCA, and DM-BCA: mass-based, rate-based, and discounted mass-based BCA. DRI: direct reduced iron.

C.4. Sensitivity Analysis

Tables C5 and C6 and Figures C2 and C3 report the sensitivity analysis (Section V.F).

Table C5. Sensitivity of Welfare and Global Emissions

Parameter		EU M-BCA		EU R-BCA		US M-BCA		US R-BCA	
		Welfare ($bn)	Abatement (%)	Welfare ($bn)	Abatement (%)	Welfare ($bn)	Abatement (%)	Welfare ($bn)	Abatement (%)
Baseline		−0.89	1.34	−1.38	1.02	−0.88	0.26	−0.32	0.06
σ_v^{Iron}	Low	−0.89	1.34	−1.38	1.02	−0.88	0.26	−0.32	0.06
σ_v^{Iron}	High	−0.89	1.34	−1.38	1.02	−0.88	0.26	−0.32	0.06
η_{Scrap}	Low	−0.83	1.29	−1.32	0.97	−0.88	0.24	−0.31	0.06
η_{Scrap}	High	−0.96	1.39	−1.44	1.07	−0.86	0.28	−0.32	0.07
σ^A_{Flat}	Low	−0.86	1.33	−1.22	1.06	−0.75	0.25	−0.31	0.06
σ^A_{Flat}	High	−0.94	1.35	−1.60	0.96	−1.07	0.28	−0.34	0.07
σ^A_{Long}	Low	−0.88	1.32	−1.31	1.02	−0.85	0.26	−0.31	0.06
σ^A_{Long}	High	−0.93	1.37	−1.48	1.01	−0.92	0.26	−0.33	0.07
$\sigma^{\text{Mat}}_{\text{EAF}}$	Low	−0.95	1.37	−1.42	1.05	−0.68	0.22	−0.32	0.07
$\sigma^{\text{Mat}}_{\text{EAF}}$	High	−0.83	1.29	−1.32	0.97	−1.09	0.29	−0.32	0.06
σ_a^{KLE}	Low	−1.50	1.29	−2.00	0.97	−1.09	0.26	−0.41	0.06
σ_a^{KLE}	High	−0.27	1.39	−0.71	1.08	−0.62	0.24	−0.21	0.06
σ_a	Low	−1.08	1.32	−1.58	0.99	−0.97	0.27	−0.36	0.07
σ_a	High	−0.61	1.37	−1.06	1.06	−0.73	0.24	−0.26	0.06

Notes: Domestic welfare (equivalent variation excluding climate damages) and global emissions reduction relative to the benchmark equilibrium, at the comparison point of Table 2. Each row multiplies one parameter by 0.5 (Low) or 2 (High), except the iron make-or-buy elasticity, whose High value is 5 times the calibrated value. Parameters: σ_v^{Iron}, iron make-or-buy; η_{Scrap}, scrap supply; σ^A_{Flat} and σ^A_{Long}, home versus imported steel; $\sigma^{\text{Mat}}_{\text{EAF}}$, material inputs of electric arc furnaces; σ_a^{KLE}, capital, labor, and energy; σ_a, top-level production nest (Table B1). M-BCA and R-BCA: mass-based and rate-based BCA.

Table C6. Sensitivity of Leakage and Pig Iron Imports

Parameter		EU M-BCA Leakage (%)	EU M-BCA Pig Iron Imports (Mt)	EU R-BCA Leakage (%)	EU R-BCA Pig Iron Imports (Mt)	US M-BCA Rev. Leakage (%)	US R-BCA Rev. Leakage (%)
Baseline		16.05	1.73	36.18	3.50	23.05	14.93
σ_v^{Iron}	Low	16.05	1.73	36.17	3.49	23.05	14.93
σ_v^{Iron}	High	16.04	1.72	36.23	3.52	23.05	14.94
η_{Scrap}	Low	19.00	1.73	39.01	3.50	24.80	16.52
η_{Scrap}	High	12.70	1.73	32.96	3.49	21.28	13.42
σ^A_{Flat}	Low	16.51	1.73	33.34	3.56	15.18	9.35
σ^A_{Flat}	High	15.21	1.72	39.96	3.41	31.83	23.44
σ^A_{Long}	Low	17.38	1.73	35.74	3.51	22.20	13.03
σ^A_{Long}	High	13.89	1.74	36.79	3.48	24.32	17.79
$\sigma^{\text{Mat}}_{\text{EAF}}$	Low	14.15	1.89	34.33	3.68	31.43	14.94
$\sigma^{\text{Mat}}_{\text{EAF}}$	High	19.17	1.54	38.86	3.22	14.68	14.38
σ_a^{KLE}	Low	18.90	1.58	39.21	3.52	24.45	18.62
σ_a^{KLE}	High	13.07	1.95	32.54	3.46	21.25	10.94
σ_a	Low	17.31	1.70	37.80	3.54	22.47	15.92
σ_a	High	14.13	1.79	33.66	3.43	24.04	13.33

Notes: Variants as in Table C5. Leakage in the EU setting, EU pig iron imports, and reverse leakage in the US setting. M-BCA and R-BCA: mass-based and rate-based BCA.

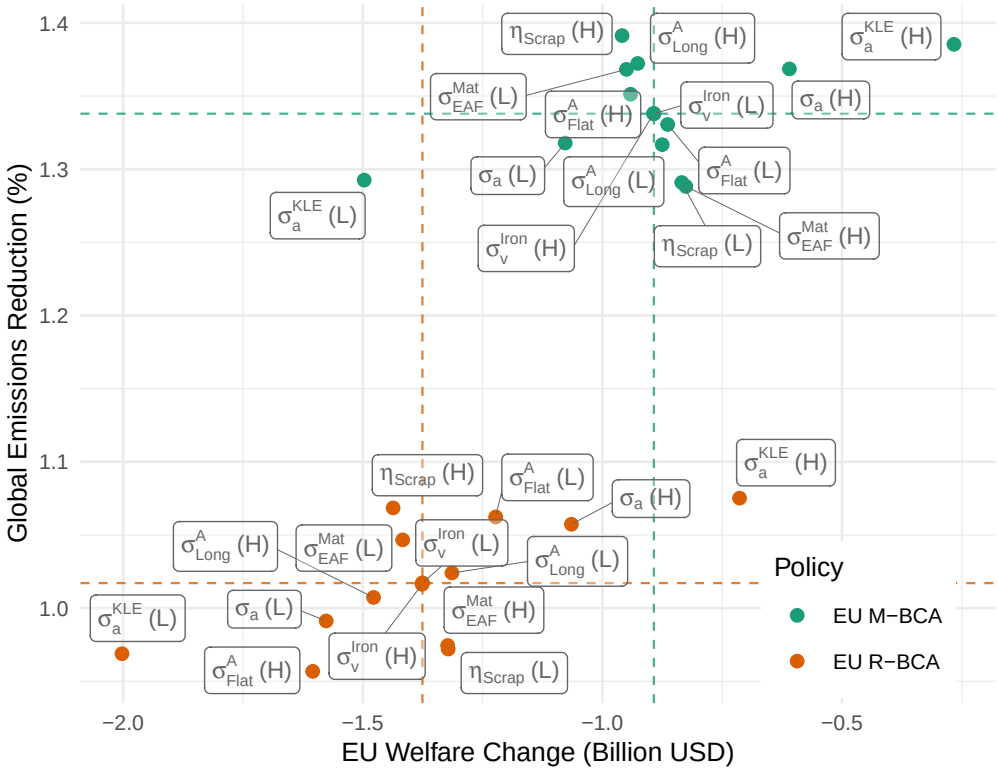


Figure C2. Sensitivity of Welfare and Global Emissions in the EU Setting

Notes: EU welfare (equivalent variation excluding climate damages) and global emissions reduction under the EU M-BCA and R-BCA for each variant in Table C5; (L) and (H) denote the Low and High values. Dashed lines mark the baseline calibration. M-BCA and R-BCA: mass-based and rate-based BCA.

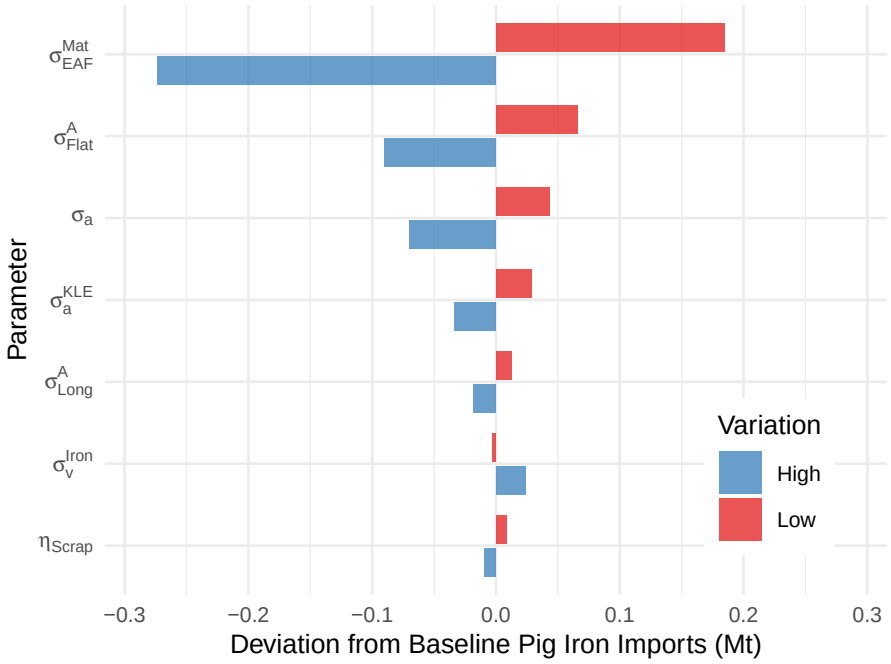


Figure C3. Sensitivity of EU Pig Iron Imports

Notes: Change in EU pig iron imports under the EU R-BCA when one parameter moves from its calibrated value to its Low or High value (Table C5), in million tonnes. R-BCA: rate-based BCA.